

Periodic Evaluation with Non-Concave Utility

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Abstract

A fund manager's performance is often evaluated annually and compared with a benchmark, such as a market index. In addition, the manager may be subject to trading constraints, such as limited use of leverage, no short-selling, and a forced liquidation clause. We formulate this as a periodic evaluation problem with a non-concave utility, a stochastic reference point, and trading constraints. The value function is characterized as the unique solution to a Hamilton-Jacobi-Bellman equation with periodic terminal and boundary conditions, which must be imposed carefully due to possible discontinuities at the terminal time and/or on the liquidation boundary. We find that, at the evaluation time, future investment opportunities induce a discontinuity in the value function on the liquidation boundary, leading to a substantial change in local risk-aversion. More importantly, this local concavity/convexity weakens and shifts inward from the liquidation boundary to the interior region as the evaluation horizon increases. As a result, the joint effect of periodic evaluation and forced liquidation can generate highly nonlinear investment strategies, which is helpful in understanding the complexity of trading strategies in the loss region.

Keywords: Periodic evaluation, Non-concave utility, Behavioral portfolio selection, HJB equation

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1 Introduction

Since the seminal work of Merton (1969), continuous-time modeling is widely used in portfolio selection, which allows for portfolio rebalancing at any time point in the investment horizon and facilitates the theoretical study. In many practical investment problems, the performance of the strategy is monitored and evaluated with a much lower frequency at a discrete set of time points. For example, index funds' performance is monitored and benchmarked daily, while the funds' rebalancing takes place intraday. Companies oversee day-to-day operations such as capital asset investments and cash management. However, the financial status is reported, and the dividends are paid only on a quarterly or even annual basis. Furthermore, individual investors may evaluate their investments periodically based on their mental accounts or tax reporting; see Benartzi and Thaler (1995) and Thaler (1999). This combination of continuous-time investment and discrete evaluation not only provides a more accurate description of practice, but also captures investors' choices in the tradeoff between the short-term performance in the current monitoring period and the long-term performance in future periods. This is crucial for understanding investors' risk-taking behavior and its dependence on the time to the next performance evaluation, since an aggressive and myopic strategy to achieve the short-term goal can hurt the performance in the long run.

In this paper, we consider a fund manager's portfolio selection problem in continuous time, whose performance is evaluated by comparing the asset under management (W) with a reference point (B) on a sequence of deterministic dates (T_i for $i = 1, 2, \dots$) such as quarterly or annually. The manager derives utility $U(W_{T_i}, B_{T_i})$ on each evaluation date T_i , and her goal is to maximize the expected total discounted utilities over an infinite horizon. Moreover, we incorporate and highlight three important features: (i) a stochastic reference point for performance evaluation, (ii) a general non-concave utility function, (iii) practical trading constraints, such as limited use of leverage, no short selling, and forced liquidation policies. These features are not only practically meaningful but also lead to novel theories as well as new economic insights.

First, the periodic evaluation of fund managers' performance based on a stochastic reference is a common practice in the fund industry, where the reference can be a stochastic benchmark such as a stock index. For example, recently, the China Securities Regulatory Commission has been considering a reduction in pay for the underperforming fund managers.¹ A stochastic reference point makes the manager allocate a certain

¹ See <https://www.bloomberg.com/news/articles/2025-03-14/china-mulls-50-pay-cuts-for-underperforming-fund-managers>

proportion of portfolio value to hedge the risk from the variations of the reference point, and therefore, it has a significant impact on the optimal investment strategies, especially when the asset under management is below the reference point and/or the evaluation time is near. Technically, introducing a stochastic reference point leads to an incomplete market, where some delicate analysis is needed. For example, as will be clear, whether or not the reference point is stochastic relates to the specification of the liquidation boundary condition.

Second, a general non-concave utility function is useful because it incorporates many interesting examples as special cases. These examples include the S -shaped utility function in behavioral economics (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992), the goal-reaching problem (Carpenter, 2000; Spivak and Cvitanić, 1999), delegated portfolio choices with non-concave compensation schemes (Carpenter, 2000), and a so-called first-loss compensation scheme where the manager uses her investment in the fund to cover any fund losses first (He and Kou, 2018). After building a general result, we focus on the fund manager’s performance evaluation problem under an S -shaped utility function. In addition, we use the homogeneity property and apply S -shaped utility to the gross return (i.e., $W/B - 1$) instead of the portfolio value (i.e., $W - B$). This formulation has been adopted in the realization utility literature; see Ingersoll and Jin (2013); He and Yang (2019); Dai et al. (2024b), and enables us to further investigate the different impacts on the optimal investment strategies from sensitivities in the gain and loss regions, respectively. This differentiation is largely neglected in the existing literature, possibly because of the models’ tractability. Compared to the one-period problem, we find that both gain and loss sensitivities have substantial impacts on the optimal investment strategies, not only in the gain region but also in the loss region. This is because the losses in the current period may be turned into gains in future periods, so the gain (loss) sensitivity may impact the strategy in the loss (gain) region in the current period.

Third, trading constraints, such as limited use of leverage, no short selling, and forced liquidation policies, are crucial in practice due to regulation. From a technical perspective, these constraints also make the optimization problem challenging to solve. Consistent with the finding in Dai et al. (2022) for a finite horizon problem, the concavification principle (i.e., replacing a non-concave utility with its concave envelope and thus reducing the non-concave utility maximization problem to a concave one) fails for the periodic problem when the trading constraints are bounded. As a result, the value function may

be discontinuous at the evaluation date. In addition, because agents may lose the future investment opportunities, they are naturally very sensitive to the liquidation policy in a periodic evaluation problem. Using a threshold liquidation policy, as commonly employed in both practice and academic literature (e.g., [Keppo and Petajisto \(2014\)](#); [He and Kou \(2018\)](#)), we find that the value function may also exhibit discontinuity at the forced liquidation boundary, which depends on the choice of reference point. If the reference point is deterministic and has a growth rate less than the risk-free rate (e.g., a constant reference point), then the manager can choose to invest only in the risk-free asset to avoid liquidation. In this case, the forced liquidation becomes an *American* option. To characterize this option, we have to impose a *Neumann*-type boundary condition, which leads to discontinuities of the value function. On the other hand, if going into liquidation is inevitable (e.g., a stochastic reference point), then the value function must be continuous on the forced liquidation boundary, where we have a *Dirichlet*-type boundary condition. To the best of our knowledge, these delicate analyses of the boundary condition are new in the literature.

From an economic perspective, in addition to competing with the reference point in the current evaluation period, the periodic evaluation leads to two new features. First, as the evaluation time is approaching, periodic evaluation can generate an extremely local risk-averse behavior (i.e., the value function is concave) and risk-seeking behavior (i.e., the value function is convex) near the liquidation boundary, if the expected value of future investments is positive and negative, respectively. Second, as the time to evaluation increases, the local risk-averse behavior and risk-seeking behavior become weak, and more importantly, these local concavity and convexity shift away from the liquidation boundary to the interior region. In particular, the local risk aversion from the periodic evaluation combined with the risk-seeking behavior from an *S*-shaped utility function can give rise to a highly non-linear investment policy in the loss region when the time to evaluation is sufficiently long: the agent may first increase then decrease and finally increase her risk exposure in the stock as the portfolio value decreases from the break-even level to the liquidation boundary. The intuition is as follows. When the portfolio value is below the reference level but far away from the liquidation boundary, the risk-seeking behavior dominates, and the agent will increase her risk exposure in the stock to gain a high expected return (and a high variance). If the portfolio value continues decreasing, by anticipating the costly liquidation, the agent will adjust her portfolio in advance for a relatively small variance (and a low return rate), which depends on the

Table 1: RELATED LITERATURE COMPARISON AMONG [DAI ET AL. \(2022\)](#), [TSE AND ZHENG \(2023\)](#) AND THIS PAPER. Here n.a. stands for not available. There are possible discontinuities at the terminal time and/or on the forced liquidation boundary. When concavification principle fails, the value function is discontinuous at the terminal/evaluation time. When Neumann type boundary condition holds, the value function is discontinuous on the forced liquidation boundary.

	Dai et al. (2022)	Tse and Zheng (2023)	This paper
<i>Panel A. Model Setup</i>			
Periodic Evaluation		✓	✓
Stochastic Reference			✓
General Non-concave	✓		✓
Trading Constraints	✓		✓
<i>Panel B. Model Solution</i>			
Method Type	Dynamic programming	Martingale duality	Dynamic programming
Terminal/Evaluation Time Condition	Continuous (non-concavify)	n.a.	Continuous (non-concavify)
	Discontinuous (concavify)		Discontinuous (concavify)
Forced Liquidation Boundary Condition	Continuous (Dirichlet)	n.a.	Continuous (Dirichlet)
			Discontinuous (Neumann)

relative expected return of the portfolio value versus the benchmark. This conservative strategy can push her portfolio away from liquidation if the evaluation time is sufficiently long. However, if the portfolio value is too low, such a conservative strategy may be inappropriate because the expected liquidation time is very short. Then, the agent will instead take a more aggressive strategy by increasing her risk exposure in the stock to gamble for large positive returns.

Related literature. Two papers are closely related to this paper. The first one is [Dai et al. \(2022\)](#), where the authors consider a finite-horizon portfolio selection problem with a general non-concave utility and a deterministic reference point. In contrast, our paper considers a periodic problem with a general stochastic reference point. For the model solution, while both papers use the dynamic programming method (e.g., [Yong and Zhou \(1999\)](#); [Fleming and Soner \(2006\)](#); [Pham \(2009\)](#)) to study an associated Hamilton-Jacobi-Bellman (HJB) equation with suitable terminal and boundary conditions, there are two major differences. On the one hand, since our problem is periodic, the terminal condition of the HJB equation contains the value function itself, which is unknown a priori. This unique feature has two important implications: (a) a fixed point must be found to solve the problem, and (b) a positive or negative jump of the value function at the liquidation

boundary, leading to substantial changes in local risk aversion and the optimal investment strategy. On the other hand, the forced liquidation boundary condition is more involved in our model due to the introduction of a general stochastic reference point. It turns out that we have two cases: Case (a) with the Dirichlet boundary condition, where the value function is continuous, and Case (b) with the Neumann boundary condition, where the value function is discontinuous. In contrast, the value function in their model is continuous on the liquidation boundary.

The other paper is [Tse and Zheng \(2023\)](#), where the authors consider a periodic problem but with a fixed reference point, an S -shaped utility function, and, in particular, no trading constraints. Consequently, their model can be viewed as a special case of our model and also needs to solve a fixed point. However, for the one-period case, they use the martingale duality method (e.g., [Berkelaar et al. \(2004\)](#); [Jin and Zhou \(2008\)](#); [He and Zhou \(2011\)](#); [Bo et al. \(2024\)](#)) to solve the problem because the market is complete, and the concavification principle holds in their model. In contrast, by incorporating a stochastic reference point, a general non-concave utility, and trading constraints, we adopt a direct dynamic programming method to establish a general result and provide further new economic insights. Table 1 provides a detailed comparison of these papers.

The combination of continuous-time portfolio management and discrete-time performance monitoring and payments has recently received attention from the literature. Besides the aforementioned [Tse and Zheng \(2023\)](#), [Keppo et al. \(2021\)](#) study a continuous-time corporate cash management problem with discrete periodic dividend payments. [Dai et al. \(2023\)](#) study the continuous-time dynamic rebalancing of leveraged exchange-traded funds where the index-tracking performance is evaluated daily. [Capponi and Zhang \(2024\)](#) develop a framework with continuous-time portfolio selection and cash withdrawals at periodic time points to fund a sequence of goals. [Dai et al. \(2024a\)](#) model the continuous-time optimal investment and consumption of an investor subject to capital gains taxes calculated at the end of each year. [Wang et al. \(2024\)](#) study an optimal portfolio problem under ratio-type periodic evaluation.

The rest of the paper is organized as follows. Section 2 presents the model. In Section 3, we solve the model using the dynamic programming principle method. Section 4 is devoted to the numerical analysis with a focus on the S -shaped utility function. Section 5 concludes. In Appendix A, we provide additional numerical results for other non-concave utility functions. Technical analysis and proofs are all delegated in Appendix B.

2 A General Periodic Evaluation Model

In this section, we lay out a general periodic evaluation model, where an agent with a non-concave utility function faces a portfolio selection problem in continuous time in each investing episode and is subject to a stochastic reference point as well as some trading constraints. We begin with the introduction of the investment opportunity.

2.1 Investment Opportunity

We consider an agent who manages an investment portfolio at any time $t \geq 0$ over an infinite horizon. The portfolio consists of a risk-free asset (i.e., bond) with a constant interest rate $r > 0$ and a risky asset (i.e., stock). The cum-dividend stock price process P_t follows a geometric Brownian motion (GBM) process:

$$dP_t = \mu P_t dt + \sigma P_t dZ_t, \quad (1)$$

where $\mu \in \mathbb{R}$ and $\sigma > 0$ are the stock's expected return rate and volatility respectively, and Z_t is a standard Brownian motion. We assume that there are no transaction costs of trading these assets.

We consider self-financing portfolio strategy represented by the proportion of portfolio value allocated to the stock π_t . Then, the portfolio value W_t evolves as

$$dW_t = (r + (\mu - r)\pi_t)W_t dt + \sigma \pi_t W_t dZ_t. \quad (2)$$

Meanwhile, the agent may also subject to some trading constraints, which will be specified shortly.

2.2 Periodic Evaluation

The agent can rebalance the portfolio continuously, but her performance is evaluated periodically on a sequence of deterministic dates such as quarterly or annually. That is, the investment horizon $[0, \infty)$ is divided into a sequence of disjoint investment episodes (T_{i-1}, T_i) with a fixed duration $T > 0$ such that

$$\cup_{i=1}^{\infty} [T_{i-1}, T_i) = [0, \infty),$$

where $T_i = iT$ for $i = 0, 1, 2, \dots$.

At the end of each investment episode (T_{i-1}, T_i) , the agent's performance is evaluated by comparing the portfolio value W_{T_i-} possibly after deducting an evaluation cost to a

reference level B_{T_i-} . This gives the agent a utility burst $U(\kappa W_{T_i-}, B_{T_i-})$, where $\kappa \in (0, 1]$ measures the cost of evaluation, i.e., $(1 - \kappa)W_{T_i-}$ is the evaluation cost. As a result, the initial portfolio value for the next investing episode becomes $W_{T_i} = \kappa W_{T_i-}$. The particular form of the utility function $U(\cdot, \cdot)$ will be specified later. We next introduce the reference point.

2.3 Reference Point

We assume that the reference B_t follows another diffusion process (e.g., a market index) during each investing episode (T_{i-1}, T_i) :

$$dB_t = \mu_B B_t dt + \sigma_B B_t d\mathcal{Z}_t^B, \quad (3)$$

where $\mu_B \in \mathbb{R}$ and $\sigma_B \geq 0$ are the expected growth rate and volatility, respectively. $\mathcal{Z}^B$ is a standard Brownian motion with a constant correlation $\rho \in [-1, 1]$ to $\mathcal{Z}$. This formulation of reference covers several examples in the literature as special cases. For example, if $\mu_B = \sigma_B = 0$, the reference point is a constant and is used in [Tse and Zheng \(2023\)](#); if $\mu_B = r$ and $\sigma_B = 0$, the reference point then grows at the risk-free rate and is adopted in [Dai et al. \(2022\)](#).

After evaluating the performance at T_i , the reference point is reset to the portfolio value for the next investing episode, i.e.,

$$B_{T_i} = W_{T_i}.$$

In other words, the reference at the beginning of the next investing episode (T_i, T_{i+1}) is normalized to the portfolio value at the end of the previous episode after the deduction.

2.4 Trading Constraints

As in practice, we impose several important trading constraints. First, we assume that the investment strategy π_t lies in a trading constraint set:

$$\pi_t \in [\underline{\pi}, \bar{\pi}], \quad (4)$$

where $-\infty \leq \underline{\pi} \leq 0 \leq \bar{\pi} \leq \infty$. For example, when $\underline{\pi} > -\infty$ and $\bar{\pi} < \infty$, the investment strategy π_t is bounded, representing a limited use of leverage.

Second, we assume that there is a liquidation event when the investment performance is below a given fraction of the reference point. To be more precise, define the liquidation

time

$$\tau_0 = \inf\{t > 0 \mid W_t \leq bB_t\} , \quad (5)$$

with the convention that $\inf\{\emptyset\} = \infty$, where $b \in [0, 1)$ is a constant. At the liquidation time τ_0 , the investor has to liquidate her position with a final utility $U(\kappa W_{\tau_0-}, B_{\tau_0-})$ and leave the market forever. Similar liquidation assumptions are also used in [Keppo and Petajisto \(2014\)](#) and [He and Kou \(2018\)](#).

2.5 Optimization Problem

With the above notations, given an initial portfolio value $W_0 = W > 0$ and reference level $B_0 = W$, the agent's optimal investment problem is to find an allocation strategy $\{\pi_t; t \geq 0\}$ to maximize her total discounted utilities

$$V_0(W) = \max_{\pi_t \in [\underline{\pi}, \bar{\pi}]} \mathbb{E}_0 \left[\sum_{i=1}^{\infty} e^{-\delta T_i} U(\kappa W_{T_i-}, B_{T_i-}) \mathbf{1}_{\{T_i < \tau_0\}} + e^{-\delta \tau_0} U(\kappa W_{\tau_0-}, B_{\tau_0-}) \right] , \quad (6)$$

subject to dynamics (2), (3), where $\delta > 0$ is a subjective discount rate, $\mathbf{1}_{\{\cdot\}}$ is an indicator function, and $\mathbb{E}_0$ is the conditional expectation on $W_0 = W$ and $B_0 = B$.

To close this section, we specify a general form utility function and provide some examples that are studied in the literature.

2.6 Functional Form of Utility

For the utility function $U(\cdot, \cdot)$, we assume that it has a lower bound²

$$U(0, B) > -\infty \quad \text{for } B \geq 0 , \quad (7)$$

and the following homogeneity property:

$$U(W, B) = B^\beta u(W/B) \quad \text{for } B > 0 , \quad (8)$$

where $\beta \in [0, 1]$ is a constant. The scaled utility function $u(\cdot)$ is increasing and has a growth rate $\alpha \leq 1$, i.e.,

$$\lim_{w \rightarrow \infty} \frac{u(w)}{w^\alpha} = C_\alpha , \quad (9)$$

for some constant $C_\alpha > 0$.³ The following are several examples of the utility functions.

² From a technical point of view, this lower bound assumption can be removed. We keep this assumption because it is sufficient for many interesting problems in application and also makes our exposition much easier.

³ This homogeneity is useful for model's dimension reduction; Our method is still applied if we remove this assumption.

Discontinuous Utility of the Goal Reaching Problem. Many portfolio managers are interested in achieving some performance goal, such as beating a stock index. [Browne \(1999\)](#) studies the optimal investment strategy of a fund manager who aims to maximize the probability of beating a benchmark by a given finite horizon, where the corresponding utility function is expressed as an indicator function

$$U(W, B) = \mathbf{1}_{\{W \geq B\}} . \quad (10)$$

Then, $\beta = 0$ and $\alpha = 0$ in this case.

Convex Utility for Option Compensation. [Carpenter \(2000\)](#) considers a risk-averse manager compensated with a call option. The manager's payoff at time T is a shares of at-the-money call option over the fund that matures at T with a strike price W_0 , which is also the reference point B .

$$U(W, B) = a \max\{W - B, 0\} . \quad (11)$$

Then, $\beta = 1$ and $\alpha = 1$ in this case.

S-shaped Utility. Following the realization utility literature ([Barberis and Xiong, 2012](#); [Ingersoll and Jin, 2013](#); [He and Yang, 2019](#); [Dai et al., 2024b](#)), the utility burst has the following form:

$$U(W, B) = B^\beta u(G/B) . \quad (12)$$

where $G = W - B$ is the gain or loss. Here the scaled function $u(\cdot)$ is a reference-level-scaled version of the Cumulative Prospect Theory (CPT) utility of [Tversky and Kahneman \(1992\)](#):

$$u(g) = \begin{cases} g^{\alpha_+} & \text{if } g \geq 0, \\ -\lambda(-g)^{\alpha_-} & \text{if } g < 0, \end{cases} \quad (13)$$

where $\lambda \geq 1$ and $\alpha_\pm \in (0, 1]$ are the three constant parameters describing $u(\cdot)$.

Additionally, $u(\cdot)$ inherits two other features of CPT: (i) the diminishing sensitivity: $u(\cdot)$ is concave ($0 < \alpha_+ \leq 1$) in the gain ($g \geq 0$) region but convex ($0 < \alpha_- \leq 1$) in the loss ($g < 0$) region, and (ii) loss aversion ($\lambda \geq 1$). A higher value of λ refers to a stronger loss aversion. Note that the $g = 0$ point is a kink where $u(\cdot)$ is not differentiable. Finally, we require:

$$\beta \leq \min\{\alpha_+, \alpha_-\}, \quad (14)$$

which ensures that $|U(G, B)|$ is decreasing in B for a fixed G . When $\beta < \min\{\alpha_+, \alpha_-\}$, the smaller the reference level, the greater the utility impact of a realized gain or loss in the absolute value $|G|$.⁴ For the special piecewise linear realization utility case used in Barberis and Xiong (2012) where $\beta = \alpha_+ = \alpha_- = 1$, $U(G, B)$ is independent of the level of B (for a given G) and does not feature the diminishing sensitivity property; see Panel A of Figure 1.

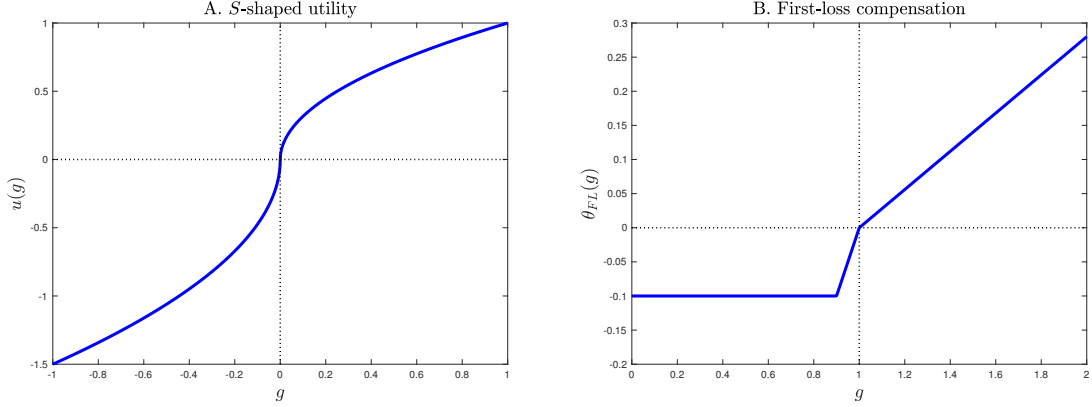


Figure 1: *S*-SHAPED SCALED UTILITY FUNCTION $u(\cdot)$ AND FIRST-LOSS SCHEME $\theta_{FL}(\cdot)$. Panel A plots the *S*-shaped scaled utility function with $\alpha_{\pm} = 0.5$, $\lambda = 1.5$. $u(\cdot)$ is convex in the loss region and concave in the gain region, and is not differentiable at the kink point $x = 0$. Panel B plots the first-loss scheme, $\theta_{FL}(\cdot)$ with $\gamma = 0.1$, $a = 0.2$.

First-Loss Scheme. He and Kou (2018) study the optimal investment strategy of a fund manager with a so-called first-loss scheme. By assuming that a proportion $\gamma \in (0, 1]$ of the fund belongs to the manager and the manager can charge a proportion $a \in [0, 1]$ of the profit, the manager’s net profit-or-loss function under the first-loss scheme is given by

$$\theta_{FL}(g) = \begin{cases} (\gamma + a(1 - \gamma))(g - 1) & \text{for } g > 1, \\ g - 1 & \text{for } 1 - \gamma \leq g < 1, \\ -\gamma & \text{for } 0 \leq g \leq 1 - \gamma. \end{cases} \quad (15)$$

Panel B of Figure 1 provides an illustration. Then the utility has the form: $U(W, B) = B^\beta u(\theta_{FL}(W/B))$, where $u(\cdot)$ satisfies (9).

⁴ Here is an illustrative example from Ingersoll and Jin (2013): “the gain or loss of \$10 is felt more strongly when the reference level is \$100 than when it is \$500.”

3 Model Solution

To solve the optimization problem (6), we use the fixed-point argument as in [Keppo et al. \(2021\)](#) and [Dai et al. \(2023\)](#). We first use the discrete-time dynamic programming principle and rewrite (6) recursively as a one-period problem. We then study the one-period problem for suitable *terminal and boundary* conditions. Finally, the original problem is solved by showing the existence and uniqueness of a fixed point.

3.1 Recursive Characterization as a One-Period Problem

For $t \in [0, T)$, we consider the following problem:

$$V(W, B, t) = \max_{\pi_t \in [\underline{\pi}, \bar{\pi}]} \mathbb{E}_t \left[\sum_{i=1}^{\infty} e^{-\delta(T_i - t)} U(\kappa W_{T_i-}, B_{T_i-}) \mathbf{1}_{\{T_i < \tau\}} + e^{-\delta(\tau - t)} U(\kappa W_{\tau-}, B_{\tau-}) \right],$$

subject to dynamics (2), (3). Here $\mathbb{E}_t$ is the conditional expectation on $W_t = W > 0$ and $B_t = B > 0$, and τ is the forced liquidation time starting from the current state

$$\tau = \inf\{s \geq t \mid W_s \leq b B_s, W_t \geq b B_t\}. \quad (16)$$

Clearly, $V_0(W) = V(W, W, 0)$.

Applying the discrete-time dynamic programming principle for infinite-horizon problems (at the discrete set of points $\{T_i\}_{i=0,1,2,\dots}$), $V(W, B, t)$ can be written recursively as the following one-period problem:

$$V(W, B, t) = \max_{\pi_t \in [\underline{\pi}, \bar{\pi}]} \mathbb{E}_t \left[e^{-\delta(T - t)} \left(U(\kappa W_{T-}, B_{T-}) + V(\kappa W_{T-}, \kappa W_{T-}, 0) \right) \mathbf{1}_{\{T < \tau\}} + e^{-\delta(\tau - t)} U(\kappa W_{\tau-}, B_{\tau-}) \mathbf{1}_{\{\tau \leq T\}} \right]. \quad (17)$$

Intuitively, if the forced liquidation doesn't happen in the first investing period, i.e., $T < \tau$, then the agent obtains a utility burst $U(\kappa W_{T-}, B_{T-})$ plus an expected value from future investments, which is given by $V(\kappa W_{T-}, \kappa W_{T-}, 0)$. Otherwise, if the forced liquidation happens before the evaluation time, i.e., $\tau \leq T$, the agent gets a utility burst $U(\kappa W_{\tau-}, B_{\tau-})$ and leaves the market forever.

The standard stochastic control theory suggests that the value function $V(W, B, t)$ is expected to solve the following HJB equation:

$$\begin{aligned} \delta V = & \frac{\partial V}{\partial t} + \frac{1}{2} \sigma_B^2 B^2 V_{BB} + \mu_B B V_B + r W V_W \\ & + \max_{\pi \in [\underline{\pi}, \bar{\pi}]} \left\{ \frac{1}{2} \sigma^2 \pi^2 W^2 V_{WW} + \rho \sigma \sigma_B \pi W B V_{WB} + (\mu - r) \pi W V_W \right\}, \end{aligned} \quad (18)$$

for $W > bB$, $B > 0$, and $t \in [0, T)$. With an abuse of notation, we use subscript to represent the partial derivative of a function, e.g., $V_W = \frac{\partial V}{\partial W}$, and $V_{WW} = \frac{\partial^2 V}{\partial W^2}$.

We next exploit the model's homogeneity to simplify the problem.

3.2 Homogeneity

Thanks to the homogeneity of the utility function, the value function is also homogeneous of degree β in W and B . We then work on the scaled state variable

$$w = \frac{W}{B} , \quad (19)$$

and the scaled value function

$$v(w, t) = B^{-\beta} V(W, B, t) . \quad (20)$$

As a result, $v(w, t)$ satisfies the following HJB equation

$$\frac{\partial v}{\partial t} + H(w, v, v_w, v_{ww}) = 0 \quad \text{for } w \in (b, \infty), t \in [0, T) , \quad (21)$$

where the Hamiltonian $H(\cdot)$ is defined as

$$\begin{aligned} H(w, p, q, A) = & \frac{1}{2} \sigma_B^2 w^2 A + \left(r - \mu_B + (1 - \beta) \sigma_B^2 \right) w q - \tilde{\delta} p \\ & + \max_{\pi \in [\underline{\pi}, \bar{\pi}]} \left\{ \left(\frac{1}{2} \sigma^2 \pi^2 - \rho \sigma \sigma_B \pi \right) w^2 A + \left(\mu - r - \rho \sigma \sigma_B (1 - \beta) \right) \pi w q \right\}, \end{aligned} \quad (22)$$

and $\tilde{\delta}$ is an effective discount rate given by

$$\tilde{\delta} = \delta - \beta \mu_B + \frac{1}{2} \sigma_B^2 \beta (1 - \beta) . \quad (23)$$

To make the value function v (or equivalently V) finite, we impose the following sufficient condition on the subjective discount rate:

$$\begin{aligned} \delta > & \mu_B \beta + (r - \mu_B) \hat{\beta} + \max \left\{ 0, \frac{1}{2} \sigma_B^2 \beta (\beta - 1) + \frac{1}{2} \sigma^2 \hat{\beta} (\hat{\beta} - 1) + \sigma_B^2 \hat{\beta} (1 - \beta) \right. \\ & \left. + \max_{\pi \in [\underline{\pi}, \bar{\pi}]} \left\{ \frac{1}{2} \sigma^2 \hat{\beta} (\hat{\beta} - 1) \pi^2 + [\mu - r - \rho \sigma \sigma_B (\hat{\beta} - \beta)] \hat{\beta} \pi \right\} \right\} , \end{aligned} \quad (24)$$

where $\hat{\beta} = \max\{\alpha, \beta\} \leq 1$. Note that when $\hat{\beta} = 1$, we have to restrict the trading set being bounded to make (24) well-defined.

3.3 Optimal Investment

The optimal strategy π^* is given by

$$\pi^* = \max \left\{ \underline{\pi}, \min \left\{ \bar{\pi}, \frac{\mu - r}{\sigma^2 \gamma_{LR}} + \rho \frac{\sigma_B}{\sigma} \left(1 - \frac{1 - \beta}{\gamma_{LR}} \right) \right\} \right\}, \quad (25)$$

where γ_{LR} is the agent's *local risk aversion* defined as

$$\gamma_{LR} = - \frac{w v_{ww}}{v_w}. \quad (26)$$

For the interior optimal investment strategy in (25), the first term $\frac{\mu - r}{\sigma^2 \gamma_{LR}}$ corresponds to the classic Merton's investment rule, while the second term $\rho \frac{\sigma_B}{\sigma} \left(1 - \frac{1 - \beta}{\gamma_{LR}} \right)$ appears because the agent needs to hedge the risk from the stochastic reference point. These optimal investment strategies are closely related to the agent's local risk aversion γ_{LR} defined in (26), which is determined by the value function's curvature.

Next, we examine the terminal condition to the HJB equation (18) or (21).

3.4 Terminal Condition

Intuitively, the definition (17) gives a heuristic condition at the terminal time $t = T$:

$$V(W, B, T) = U(\kappa W, B) + V(\kappa W, \kappa W, 0) \mathbf{1}_{\{W > bB\}}, \quad (27)$$

or equivalently,

$$v(w, T) = F(w) := u(\kappa w) + (\kappa w)^\beta v(1, 0) \mathbf{1}_{\{w > b\}}. \quad (28)$$

Note that the terminal condition (28) contains two parts: (i) a utility burst $u(\kappa w)$ from the current evaluation, and (ii) a continuation value $(\kappa w)^\beta v(1, 0) \mathbf{1}_{\{w > b\}}$ from the future evaluation, which depends on the value function itself and therefore shows the periodic nature of this problem. In addition, there is a jump on the forced liquidation boundary $w = b$, which makes the local risk attitude extremely large in magnitude (either risk-averse or risk-seeking). More importantly, this local behavior will transmit into the interior region as will be clearly illustrated in our numerical analysis in Section 4.

However, in the presence of trading constraints, the proposed terminal condition (27) or (28) may not be appropriate. In fact, for a non-periodic problem, Dai et al. (2022) have shown that the “true” terminal condition depends on whether the trading constraint set is bounded or not. In Appendix B, we will show that their results still hold in our

periodic problem. So, instead of using (28), we impose the following terminal condition for v :

$$\begin{cases} \lim_{(z,t) \rightarrow (w,T-)} v(z,t) - F(w-) \\ \quad - 2\Phi\left(\frac{\min\{0, \ln(z/w)\}}{L\sqrt{T-t}}\right) (F(w) - F(w-)) = 0 & \text{if } [\underline{\pi}, \bar{\pi}] \text{ is bounded ,} \\ \lim_{(z,t) \rightarrow (w,T-)} v(z,t) = \widehat{F}(w) & \text{if } [\underline{\pi}, \bar{\pi}] \text{ is unbounded ,} \end{cases} \quad (29)$$

where constant $L = \sigma \max\{|\underline{\pi}|, |\bar{\pi}|\}$, $\Phi(\cdot)$ is the standard normal cumulative distribution function, and $\widehat{F}(\cdot)$ is the concave envelope of function $F(\cdot)$.

The intuition is the same as in Dai et al. (2022). When the trading constraint set is unbounded, the agent can always exploit unlimited leverage to gamble on favorable outcomes, which lifts the value function at the terminal time to its concave envelope. In contrast, this extreme gambling strategy is ruled out when the trading set is bounded.

Now let us consider the condition on the forced liquidation boundary.

3.5 Boundary Condition

Due to trading constraints, the definition of (16) would suggest the following heuristic condition on the forced liquidation boundary

$$V(W, B, t) = U(\kappa W, B) \quad \text{when } W = bB, t \in [0, T] , \quad (30)$$

or equivalently,

$$v(b, t) = u(\kappa b) . \quad (31)$$

Unfortunately, this boundary condition may also be invalid in general. To see this, consider the case where $\mu_B = r$ and $\sigma_B = 0$, and assume that $v(1, 0) > 0$ and $u(\kappa b) < 0$. In Appendix B.1 we have shown that the stochastic control problem (17) is equivalent to that in (34). So, taking $\pi_t \equiv 0$ in (35) leads to $dw_t \equiv 0$, which implies that the agent can stay in the solvency region forever. From the definition (34), we immediately have, for $t \in [0, T)$,

$$\lim_{w \rightarrow b+} v(w, t) \geq \lim_{w \rightarrow b+} e^{-\tilde{\delta}(T-t)} F(w) > u(\kappa b) + e^{-\tilde{\delta}(T-t)} v(1, 0) (\kappa b)^\beta > u(\kappa b) ,$$

which shows that the boundary condition (31) is inappropriate. The key point here is that whether or not to liquidate becomes an American option, if the probability of going bankrupt near the liquidation boundary $w = b$ is zero under some admissible strategy. As a consequence, we divide the boundary condition into two cases:

Case (a): $\sigma_B > 0$ or $\mu_B > r$. When the probability of going bankrupt is strictly positive for all admissible strategies, the Dirichlet-type boundary condition (31) holds.

Case (b): $\sigma_B = 0$ and $\mu_B \leq r$. When the probability of going bankrupt is zero for some admissible strategy ($\pi_t = 0$), the Neumann-type boundary condition holds:

$$\min \left\{ -\frac{\partial v}{\partial t} - (r - \mu_B)b v_w + \tilde{\delta}v, v - u(\kappa b) \right\} = 0 \quad \text{for } t \in [0, T) . \quad (32)$$

Equation (32) is a variational inequality, which is similar to that satisfied by an American option. If the forced liquidation value $u(\kappa b)$ is greater than the continuation value, liquidation is optimal by choosing $\pi_t \neq 0$ and this yields $v = u(\kappa b)$. In contrast, if $v > u(\kappa b)$, then staying alive is optimal by setting $\pi_t = 0$, which leads to the equation that $-v_t - (r - \mu_B)b v_w + \tilde{\delta}v = 0$. Combining these two actions, we obtain the Neumann boundary condition (32). However, a key difference is that there is no *diffusion* term in (32), which is necessary to keep the agent away from forced liquidation. In addition, note that the above reasoning is also true at the evaluation time, so the terminal condition at the corner point (b, T) should be

$$\lim_{(z,t) \rightarrow (b+, T-)} v(z, t) = \max\{F(b+), u(\kappa b)\} . \quad (33)$$

In particular, for the one-period problem studied in Dai et al. (2022), where $\delta = 0$, $\beta = 0$, $\mu_B = r$, $\kappa = 1$, and thereby $F(b+) = u(b)$, equations (32) and (33) has a closed-form solution $\phi(b, t) = u(b)$, which is a Dirichlet-type condition, too.

The following theorem summarizes the above discussions.

Theorem 1 (Existence and Uniqueness). *Let the condition (24) hold. There exists a unique viscosity solution $v \in C([b, \infty) \times [0, T))$ to the HJB equation (21) coupled with the terminal condition (29) and Dirichlet boundary condition (31) for Case (a) or coupled with the terminal condition (29) and (33) and Neumann boundary condition (32) for Case (b). In addition, if the reference is deterministic $\sigma_B = 0$ and the trading set is unbounded, then $v \in C^{2,1}((b, \infty) \times [0, T))$.*

The proof of Theorem 1 and related theoretical analysis are deferred to Appendix B. We next turn to an extensive numerical analysis.

4 Numerical Analysis

In this section, we conduct an extensive numerical analysis based on our general model. To illustrate the interaction between the periodicity and nonconcavity of the problem,

we focus on the case of an S -shaped utility function given in (12) and (13), which demonstrates rich economic insights.

4.1 Baseline Parameter Values

The analysis is based on the following baseline parameters. First, for the investment opportunity, we assume that the risk-free rate is 3% per year ($r = 0.03$), and that the stock's expected return rate and volatility are 9% ($\mu = 0.09$) and 30% ($\sigma = 0.3$) per year, respectively. Second, for the reference point, we shall use a market index and assume that its annual expected return rate and volatility are 7% ($\mu_B = 0.07$) and 10% ($\sigma_B = 0.1$), respectively. The correlation between the market index and the stock is assumed to be 0.5 ($\rho = 0.5$). With these values, the market price of risk is fixed at $\lambda_M = 0.4$, which is broadly consistent with the empirical studies in the U.S. stock market. Third, for the S -shaped utility function, following Dai et al. (2024b) we set the gain and loss sensitivity parameters both at 0.5 ($\alpha_+ = \alpha_- = 0.5$), the loss aversion parameter at $\lambda = 1.5$, and the scaling parameter $\beta = 0.3$. Fourth, for the trading constraints, we intentionally choose non-symmetric trading bounds, where the ratio of stock investment to the portfolio value lies within the interval $[\underline{\pi}, \bar{\pi}] = [-5, 3]$. Moreover, forced liquidation occurs once the portfolio value is below 50% ($b = 0.5$) of the reference level. Next, we assume that there is a 2% evaluation cost proportional to the portfolio value at the evaluation time. As annual assessments are commonly adopted in practice, we set $T = 1$. Finally, the subjective discount rate is fixed at 8% ($\delta = 0.08$), which is sufficient to make the value function finite. For easy reference, we summarize our baseline parameter values in Table 2.

Table 2: Baseline Parameter Values

Bond and stock:	$r = 0.03$	$\mu = 0.09$	$\sigma = 0.3$
Market reference:	$\mu_B = 0.07$	$\sigma_B = 0.1$	$\rho = 0.5$
S -shaped utility:	$\alpha_+ = 0.5$	$\alpha_- = 0.5$	$\lambda = 1.5 \quad \beta = 0.3$
Trading constraints:	$\underline{\pi} = -5$	$\bar{\pi} = 3$	$b = 0.5$
Other parameters:	$\kappa = 0.98$	$\delta = 0.08$	$T = 1$

4.2 Non-concavity and the Value Function

We first investigate the value function. Figure 2 plots the scaled value function $v(w, t)$ when t approaches the evaluation time T and when w decreases to the liquidation boundary b for the baseline parameter values in Table 2. It turns out that the scaled value function at inception $v(1, 0) = 0.37 > 0$, which implies that the total expected discounted utility from the periodic evaluation is positive. With this value of $v(1, 0)$, we plot the terminal value at $t = T$ (the dashed blue line) and the value at $t = 0.999$ (the solid red line) for different portfolio value levels in Panel A. These two curves turn out to be very close, which demonstrates that the value function is continuous at the evaluation time $t = T$ as $t \rightarrow T^-$ when the trading constraint is bounded (i.e., $[\underline{\pi}, \bar{\pi}] = [-5, 3]$), consistent with our theoretical result.

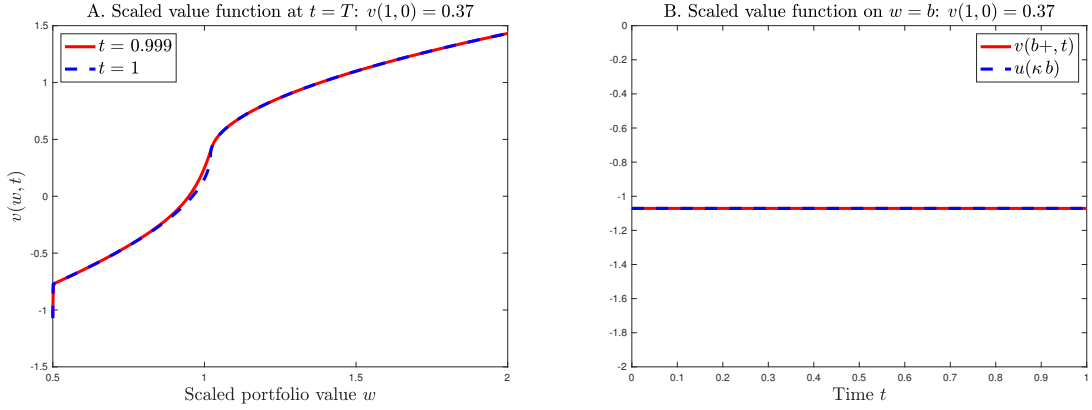


Figure 2: CONTINUITY OF SCALED VALUE FUNCTION v . Panel A illustrates the continuity of the value function $v(w, t)$ as t approaches the evaluation time T . Panel B illustrates the continuity of the value function $v(w, t)$ as w approaches the liquidation boundary b .

On the other hand, due to the randomness of the reference point ($\sigma_B > 0$), the value function should be continuous on the liquidation boundary $w = b$ (i.e., $\lim_{w \rightarrow b+} v(w, t) = u(\kappa b)$) as predicted in Case (a) with Dirichlet boundary condition (31). This is verified in Panel B of Figure 2, where $v(b+, t)$ (the solid red line) and $u(\kappa b)$ (the dashed blue line) almost coincide with each other.

Another interesting observation is that the terminal value $v(w, T)$ has a positive jump on the liquidation boundary $w = b$. This can be attributed to the fact that when w is even slightly above the liquidation boundary b , the agent's investment survives and is carried forward into the next period, and therefore it can benefit from the positive expected utility ($v(1, 0) > 0$) for the future investments. However, when $w = b$, the investment portfolio has to be immediately liquidated without future opportunities. This

makes the agent extremely risk-averse near the liquidation boundary. It is worth pointing out that this local risk aversion will transmit into the interior region rather than stay at the liquidation boundary as the evaluation time becomes remote. Now, let us investigate the optimal investment strategy.

4.3 Impact of Periodic Evaluation on the Optimal Investment Strategy

Next, we illustrate the impact of the periodic evaluation on the investment strategy by comparing it with the one-period problem.

Panel A of Figure 3 illustrates the optimal investment strategy in the loss region in the one-period case. Similar to Dai et al. (2022), the optimal strategy exhibits a risk-taking behavior: The investor will gamble for the portfolio value to beat the benchmark (that is, w to be above 1) by taking the maximum leverage. Taking $t = 0.99$ as an example (the red solid line in Panel A), when the portfolio value w is moderately below 1 (e.g., $w \geq 0.54$), the investor maximizes the leverage by taking $\pi = -5$. As the portfolio value decreases towards the liquidation boundary $b = 0.5$, the investor will put more emphasis on avoiding liquidation, by first switching to a positive maximum leverage $\pi = 3$ (due to the positive risk premium), and then gradually reducing π when w becomes lower than 0.51.

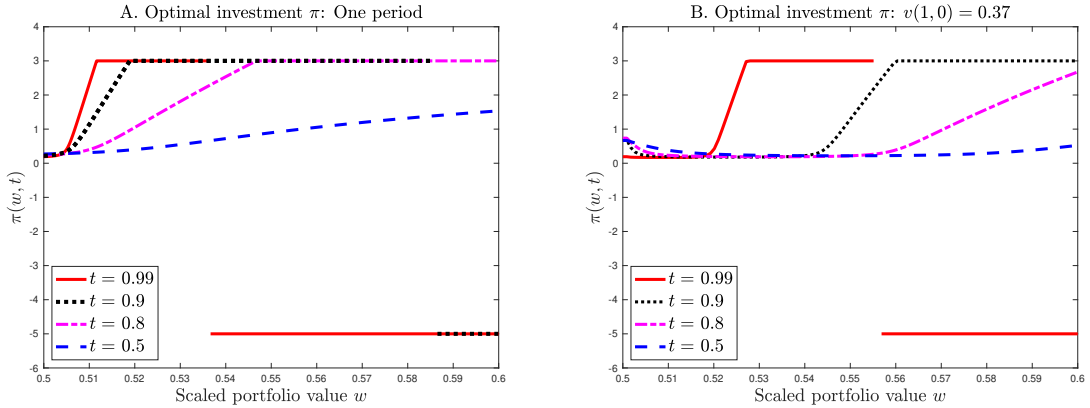


Figure 3: OPTIMAL INVESTMENT IN THE LOSS REGION. Panel A plots the optimal investment strategy $\pi(w, t)$ for the one-period case, and Panel B for the periodic case. π is non-monotonic as w_t decreases to the liquidation boundary $w = b = 0.5$.

In stark contrast, the periodic evaluation case exhibits a new risk-taking behavior: when $v(1, 0)$ is sufficiently positive, the investor has an additional incentive to gamble to avoid liquidation when the maturity is not imminent. Indeed, taking again $t = 0.99$ as an example, Panel B of Figure 3 shows that the investor will increase π when w is below

0.51. On the other hand, the investor will act more conservatively at a moderate loss level, by starting to reduce leverage in advance when w drops below 0.56 (compared to 0.54 in the one-period case).

These observations can be interpreted as follows. By anticipating the costly liquidation, the agent will adjust her portfolio in advance for a relatively small variance (and a low return rate), which depends on the relative expected return of portfolio value versus the benchmark. This conservative strategy can push her portfolio away from the liquidation if the evaluation horizon is sufficiently long. However, if the agent's portfolio value is too low, such a conservative strategy may be inappropriate. Then the agent will instead take a more aggressive strategy by increasing her risk exposure in the stock to gamble for positive returns. This explains the decrease-then-increase pattern of the optimal investment when the evaluation time is in distance.

This is also confirmed by the local risk aversion $\gamma_{LR}(w, t)$ in Figure 4. In the one-period case in Panel A, the investor's risk aversion maintains a very low level when w is moderately below the benchmark and significantly increases when w further drops (e.g., below 0.505). In contrast, in the periodic case as shown in Panel B, the risk aversion significantly increases when w drops below 0.52, but as w keeps decreasing towards the liquidation boundary b , the risk aversion will significantly decrease again. In other words, the investor's low risk aversion when $w \in [0.52, 1]$ reflects the incentive to gamble to beat the benchmark, and her low risk aversion when $w \in [0.5, 0.51]$ reflects the incentive to gamble to avoid liquidation.

Moreover, when the evaluation time becomes distant, γ_{LR} has a lower peak and, more importantly, its peak shifts away from the liquidation boundary. In other words, the local concavity arising from the positive jump of the terminal value on the liquidation boundary tends to transmit into the interior region as the time to evaluation increases.

4.4 Impact of Sensitivities in Gain and Loss Regions: $\alpha_{\pm}$

Many existing papers assume that the gain and loss sensitivities α_+ and α_- are the same, partly for model tractability. In this subsection, we study the impact of the asymmetry in these two parameters.

One-Period Case. To understand the impact of our periodic problem, we first illustrate the optimal investment strategies in the corresponding one-period problem, plotted in Figure 5. This figure shows two implications. First, given α_+ and α_- , when

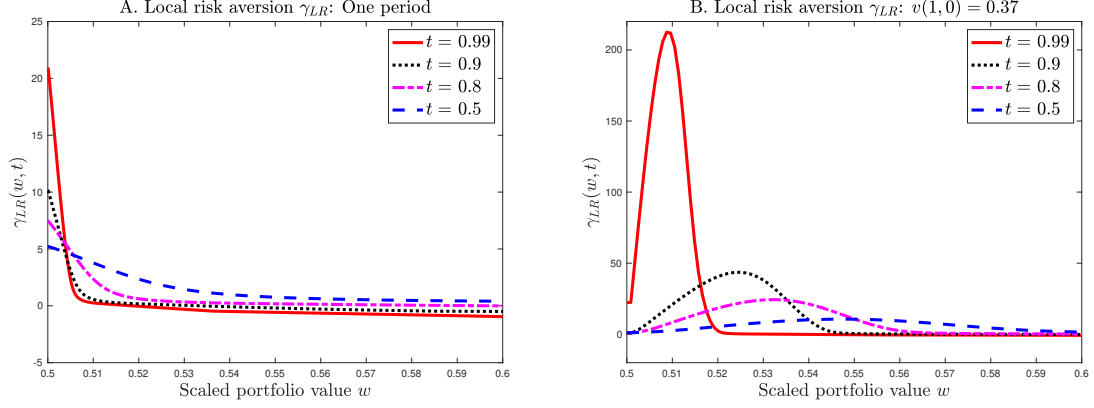


Figure 4: LOCAL RISK AVERSION IN THE LOSS REGION. Panel A plots the local risk aversion $\gamma_{LR}(w, t)$ for the one-period case, and Panel B for the periodic case. In the periodic case, as the evaluation time becomes distant, the peak of γ_{LR} decreases and shifts away from b to the interior region.

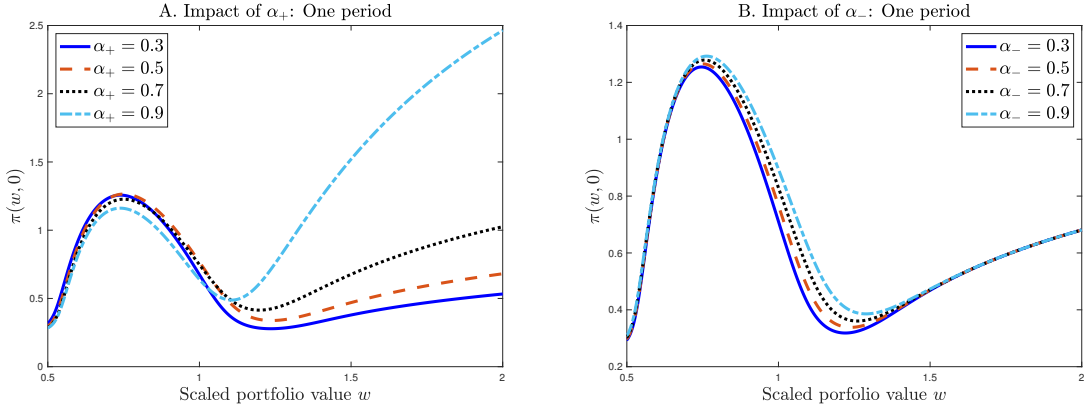


Figure 5: OPTIMAL INVESTMENT STRATEGY $\pi(w, 0)$ WITH DIFFERENT SENSITIVITIES IN GAIN AND LOSS REGIONS FOR ONE-PERIOD PROBLEM. Panels A and B plot the optimal investment strategy $\pi(w, 0)$ for different sensitivity α_+ in the gain region and α_- in the loss region, respectively.

the scaled portfolio value w is sufficiently large (e.g., $w \geq 1.3$ in Figure 5), the optimal investment $\pi(w, 0)$ increases with w . However, as w decreases into the loss region and towards the liquidation boundary b , the optimal investment $\pi(w, 0)$ first increases and then decreases. This is because the convexity of the S-shaped utility in the loss region makes the investor's local risk aversion $\gamma_{LR}(w, 0)$ first decrease and then increase as w decreases toward the liquidation boundary. In other words, the investor first increases the investment in the stock to gamble for a positive return, but as w decreases to a sufficiently low level, the investor will disinvest in the stock to avoid liquidation. Second, the change in the gain sensitivity α_+ mainly impacts the investment in the gain region (e.g., $w \geq 1.3$), where a larger α_+ means that the investor becomes less risk-averse and therefore puts more portfolio value into the stock in the gain region. In contrast, the change in the loss sensitivity α_- mainly impacts the investment around the break-even point $w = 1$.⁵

Now we turn to our periodic problem.

Periodic Case. Panels A1 and B1 of Figure 6 plot the scaled value function $v(1, 0)$ with respect to α_+ and α_- , respectively. $v(1, 0)$ decreases from about 2.3 to -0.5 as α_+ increases from 0.3 to 1. In contrast, $v(1, 0)$ increases from the level close to 0 to about 1.3 as α_- increases from 0.3 to 1. This is because a smaller α_+ leads to a higher marginal utility, which in turn enhances the total utility obtained. Similarly, a smaller α_- results in a larger marginal disutility, which in turn decreases the total utilities.

Meanwhile, Panels A2 and B2 plot the optimal investment strategies $\pi(w, 0)$ for different α_+ and α_- , respectively. These two panels unveil two insights from the periodic problem that significantly differ from those in the one-period problem. First, α_+ has a big impact on the optimal investment π not only in the gain region but also in the loss region, as illustrated in Panel A2 for w between 0 and 1. Indeed, in a periodic problem, the losses in the current period may be turned into gains in future periods, so that the gains sensitivity may impact the strategy in the loss region in the current period. Similarly, α_- now also has a bigger impact on the investment in the gain region compared to the one-period problem, as illustrated in Panel B2. However, such an impact is only moderate. In fact, one can show that as the portfolio value level goes to infinity, the

⁵ To be more precise, with the evaluation cost ($\kappa = 2\%$), the break-even point should be at $w = 1/(1 - \kappa) \approx 1.02$. For easy presentation, we simply use $w = 1$ as the break-even point.

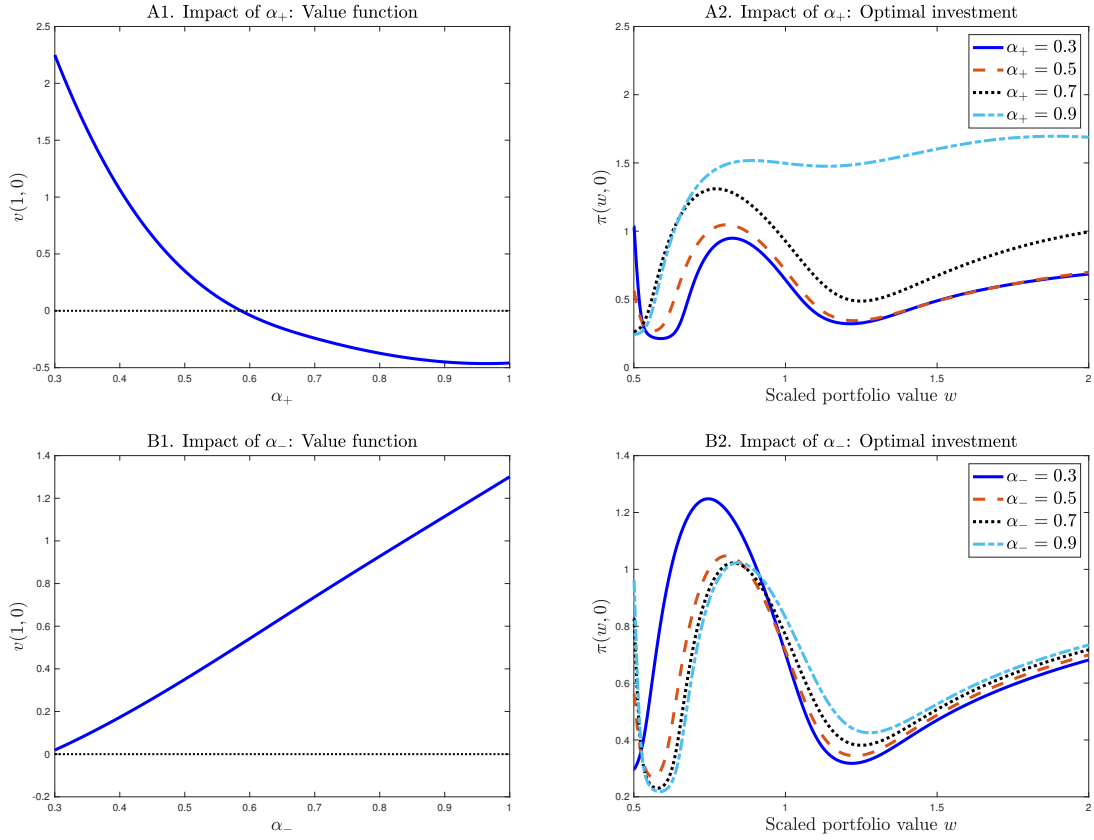


Figure 6: SCALED VALUE FUNCTION $v(1, 0)$ AND OPTIMAL INVESTMENT STRATEGY $\pi(w, 0)$ WITH DIFFERENT SENSITIVITIES IN GAIN AND LOSS REGIONS Panel A1 (B1) plots the value function $v(1, 0)$, and Panel A2 (B2) plots the optimal investment strategy $\pi(w, 0)$ for different sensitivity α_+ in gain region (α_- in loss region). Other parameter values are listed in Table 2.

optimal investment strategy will converge to the Merton line with a hedge-term

$$\lim_{w \rightarrow \infty} \pi(w, 0) = \frac{\mu - r}{\sigma^2(1 - \alpha_+)} + \rho \frac{\sigma_B}{\sigma} \left(1 - \frac{1 - \beta}{1 - \alpha_+} \right),$$

which is independent of α_- .

Second, from Panel A2 of Figure 6, when α_+ is small (e.g., $\alpha_+ = 0.3$ and $\alpha_+ = 0.5$), the optimal investment $\pi(w, 0)$ will first increase then decrease then finally increase as w_t decreases from the break even point ($w = 1$) to the forced liquidation boundary b , and this last increase in investment right before reaching the liquidation boundary is in stark contrast to the one-period problem. This increase can be interpreted through the discussion for Figure 4. Indeed, in this case, $v(1, 0)$ is sufficiently positive, leading to a significant local concavity on the liquidation boundary when the time to evaluation is short; see Panel A of Figure 2 for illustration. As we discussed earlier, this concavity will transmit into the interior region as the time to evaluation increases. As a result, this concavity combined with the convexity of the S -shaped utility in the loss region makes the investor's local risk aversion $\gamma_{LR}(w, 0)$ decrease first then increase and finally decrease as w drops from the break even point ($w = 1$) to the liquidation boundary b . Put differently, this transition of the local concavity gives the investor a strong incentive to dis-invest in the stock to avoid liquidation much earlier before hitting the liquidation boundary. But, if the portfolio value level continues dropping to a sufficiently low level, then the expected liquidation time is near, and the investor is likely to take a bet by putting more portfolio value in the risky asset to gain a higher chance of getting away from the liquidation.

In contrast, for large values of α_+ (e.g., $\alpha_+ = 0.7$ and $\alpha_+ = 0.9$), $v(1, 0)$ becomes negative, leading to a local convexity on the liquidation boundary when the time to evaluation is short. Again, this local convexity will transmit into the interior region as the time to evaluation becomes distant. This convexity, combined with the convexity from the S -shaped utility, makes the investor invest more in the risky asset when the loss is moderate. As the loss keeps growing, the investor dis-invests in the risky asset because the risk-seeking behavior also decreases.

We next examine the impact of the reference point.

4.5 Impact of Reference Points

In most existing literature, reference points are simply taken to be deterministic. However, the reference points in mutual funds are typically stochastic. In this subsection, we

study how reference points affect the optimal investment strategies. We first consider the correlation between the stock and the reference point.

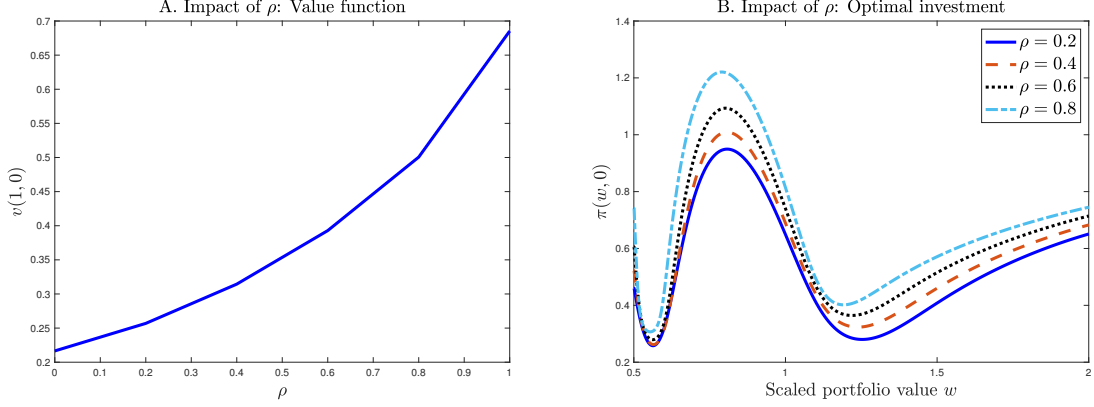


Figure 7: SCALED VALUE FUNCTION $v(1,0)$ AND OPTIMAL INVESTMENT STRATEGY $\pi(w,0)$ WITH DIFFERENT CORRELATIONS OF REFERENCE POINT. Panel A plots the value function $v(1,0)$, and Panel A plots the optimal investment strategy $\pi(w,0)$ for different correlations ρ . Other parameter values are listed in Table 2.

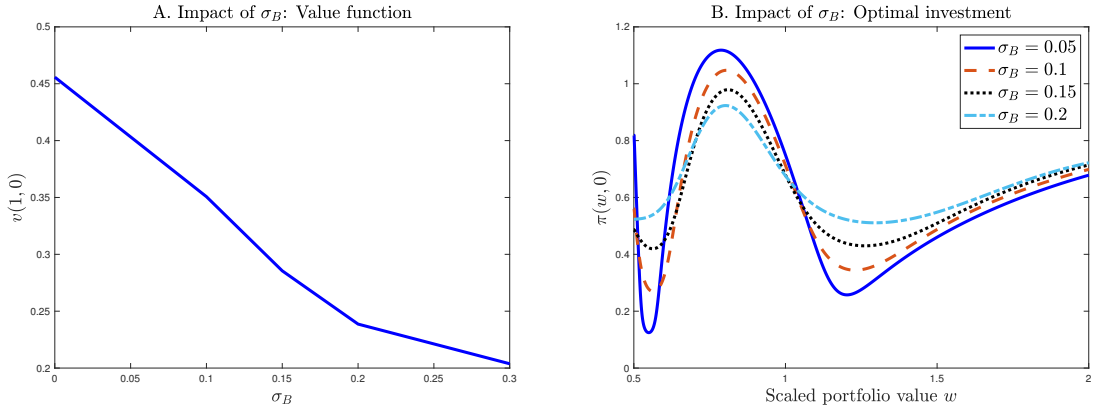


Figure 8: SCALED VALUE FUNCTION $v(1,0)$ AND OPTIMAL INVESTMENT STRATEGY $\pi(w,0)$ WITH DIFFERENT VOLATILITIES OF REFERENCE POINT. Panel A plots the value function $v(1,0)$, and Panel A plots the optimal investment strategy $\pi(w,0)$ for different volatilities σ_B . Other parameter values are listed in Table 2.

Correlation ρ between Stock and Reference. Using a competitor's performance as a benchmark is commonly seen in practice; we study this interesting problem in a parsimonious way by examining the impact of correlation ρ in the range of $[0, 1]$. In Figure 7, Panel A plots the value function $v(1,0)$, and Panel A plots the optimal investment strategy $\pi(w,0)$ for different correlations ρ . Other parameter values are retained and documented in Table 2. The monotonic pattern in both Panels A and

B indicates that, as the correlation ρ increases, the agent will take a more aggressive strategy $\pi(w, 0)$ to beat the reference point, which enhances her expected value $v(1, 0)$.

Reference's Uncertainty σ_B . Figure 8 examines the impact of σ_B , where Panel A plots the value function $v(1, 0)$, and Panel B plots the optimal investment strategy $\pi(w, 0)$ for different volatilities σ_B . Other parameter values are fixed and summarized in Table 2. Naturally, as the uncertainty of the reference point increases, the expected value decreases, and the agent should take a more conservative investment strategy in the loss region to reduce the probability of costly liquidation. This is consistent with the empirical observations that investments are reduced during bear markets with high uncertainties.

Deterministic Reference Point Next, we turn to the case of a deterministic reference point by setting $\sigma_B = 0$. Panel A1 of Figure 9 shows that the value function is discontinuous in t on the liquidation boundary, i.e., $\lim_{w \rightarrow b+} v(w, t) > u(\kappa b)$. Panel A2 further indicates that the optimal investment at $w = b+$ is to invest only in the risk-free asset (i.e., $\lim_{w \rightarrow b+} \pi(w, t) = 0$). Consequently, we have the Neumann type boundary condition (44) in this case. The intuition is that since the agent can choose not to invest in the risky asset to avoid liquidation, the forced liquidation essentially becomes an American option. If liquidation is strictly suboptimal, as in our numerical example, the investor will take the most conservative strategy by putting all portfolio value in the risk-free asset.

However, it is worth pointing out that in some circumstances, which heavily depend on the sign of $v(1, 0)$, it may be optimal to liquidate before the evaluation time. Panels B1 and B2 in Figure 9 provide such an example where $\sigma_B = 0$, $\mu_B = 0 < r$, and a 50% evaluation cost $\kappa = 0.5$, which leads to a negative expected value $v(1, 0) = -1.16 < 0$. In this particular scenario, it turns out that the optimal liquidation time is around $t = 0.99$, after which the value function $v(b, t)$ and the liquidation value $u(\kappa b)$ coincide.

4.6 Unbounded Trading Constraint

Now let us relax the trading constraint by setting $\underline{\pi} = -\infty$ and $\bar{\pi} = \infty$. In Figure 10, Panel A shows that the value function converges to the concave envelope of the terminal condition $\hat{F}(\cdot)$ and is therefore not continuous at the evaluation time. This is consistent with the theoretical result (29). Panel B further demonstrates that the optimal

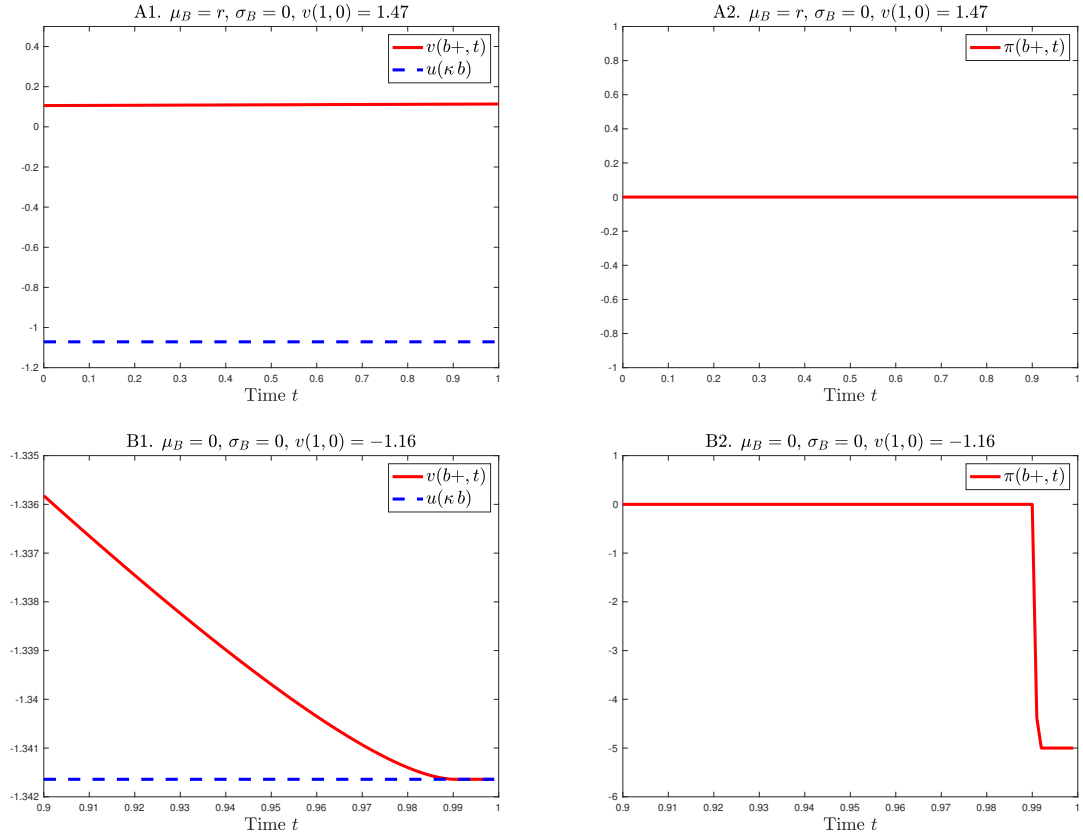


Figure 9: SCALED VALUE FUNCTION v AND OPTIMAL INVESTMENT STRATEGY π : DISCONTINUITY ON THE LIQUIDATION BOUNDARY. Panel A1 (B1) plots the value function v , and Panel A2 (B2) plots the optimal investment strategy π when $\kappa = 0.98$, $\mu_B = r$, $v(1,0) = 1.47 > 0$ (when $\kappa = 0.5$, $\mu_B = 0 < r$, $v(1,0) = -1.16 < 0$). Other parameter values are listed in Table 2.

investment strategy is extremely large in the loss region when the time to evaluation is approaching. The intuition is that, with an unbounded trading set, the investor can take a significant bet on the risky asset to beat the reference when the evaluation time is close. By employing such a strategy, the scaled value function at inception $v(1, 0) = 0.61$, which is higher than 0.37 in our baseline case.

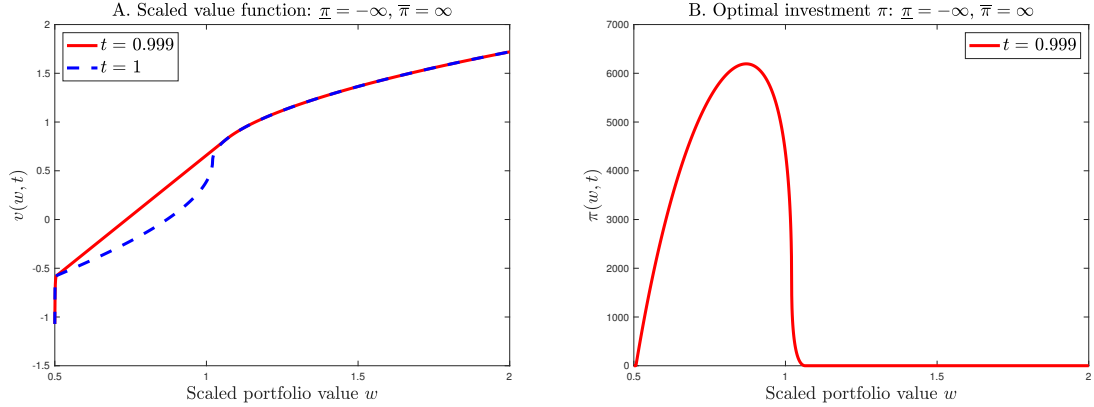


Figure 10: SCALED VALUE FUNCTION v AND OPTIMAL INVESTMENT STRATEGY π WITH AN UNBOUNDED TRADING SET: $\underline{\pi} = -\infty$ AND $\bar{\pi} = \infty$. Panel A plots the value function v . Panel B plots the optimal investment strategy π . The unbounded trading set yields a higher scaled value at inception than that in our baseline case: $v(1, 0) = 0.61 > 0.37$. Other parameter values are listed in Table 2.

5 Conclusion

Periodic evaluation problems are common in practice. In this paper, we propose a model with a general non-concave utility, a stochastic reference point, and trading constraints, such as limited use of leverage, no short-selling, and a forced liquidation clause. Methodologically, we employ dynamic programming to characterize the value function as the unique viscosity solution to an HJB equation with periodic terminal and boundary conditions, which must be imposed carefully due to possible discontinuities at the terminal time and/or on the liquidation boundary. For the terminal condition, we extend the result in Dai et al. (2022) to our periodic problem, where the concavification principle fails when the trading constraint is bounded. On the liquidation boundary, we have a Dirichlet condition for Case (a) where the value function is continuous and Neumann condition for Case (b) where the value function is discontinuous.

Regarding the model's implication, we have two main findings. First, as the evaluation time is approaching, periodic evaluation can generate an extremely local risk-averse behavior (i.e., the value function is concave) and risk-seeking behavior (i.e., the value

function is convex) near the liquidation boundary, if the expected value of future investments is positive and negative, respectively. Second, as the time to evaluation increases, the local risk-averse behavior and risk-seeking behavior become weak, and more importantly, these local concavity and convexity shift inward from the liquidation boundary to the interior region. In particular, the local risk aversion from the periodic evaluation combined with the risk-seeking behavior from an S -shaped utility function can give rise to a highly non-linear investment policy in the loss region. This is helpful for us in understanding the complexity of investor trading behaviors under loss conditions.

A Additional Numerical Results

In this appendix, we show that our numerical results presented for the S -shaped utility function in Section 4 are robust for other non-concave utility functions. To be more precise, we consider the following three utility functions: 1) Discontinuous utility for goal reaching problem; 2) Convex utility for option compensation; 3) First-loss scheme in fund management.

Figure 11 plots the scaled value function $v(w, t)$, local risk aversion $\gamma_{LR}(w, t)$ and optimal investment $\pi(w, t)$ for these three non-concave utility functions. In particular, for the goal reaching problem, we intentionally assume a large evaluation cost, $\kappa = 0.6$. For option compensation, due to the linear growth rate, we use a large discount rate $\delta = 0.4$. For the first-loss scheme, we assume that the proportions $\gamma = 0.1$ and $a = 0.2$ in (15), and the scaled utility $u(\cdot)$ is an S -shape given by (13). Other parameter values are listed in Table 2. Overall, the results for the S -shaped utility function analyzed in Section 4 still hold for these three non-concave utility functions. In particular, Panels A1, B1, and C1 plot the scaled value function $v(w, t)$ when the time t is approaching the evaluation time T . Due to the bounded trading set, the value function is continuous at the evaluation time. Panels A2, B2, and C2 illustrate the local risk aversion $\gamma_{LR}(w, t)$ with different evaluation horizons for three cases. Because the expected value $v(1, 0)$ is sufficiently positive, the local risk aversions are extremely large for short evaluation horizon and decay fast as the evaluation horizon increases. Meanwhile, their peak values shift away from the liquidation boundary to the interior region. Finally, Panels A3, B3, and C3 show that the optimal investment strategies are highly non-linear in the loss region. Different from the one-period case, the agent increases her risk exposure in the

stock to make gambles when the portfolio value is sufficiently close to the liquidation boundary.

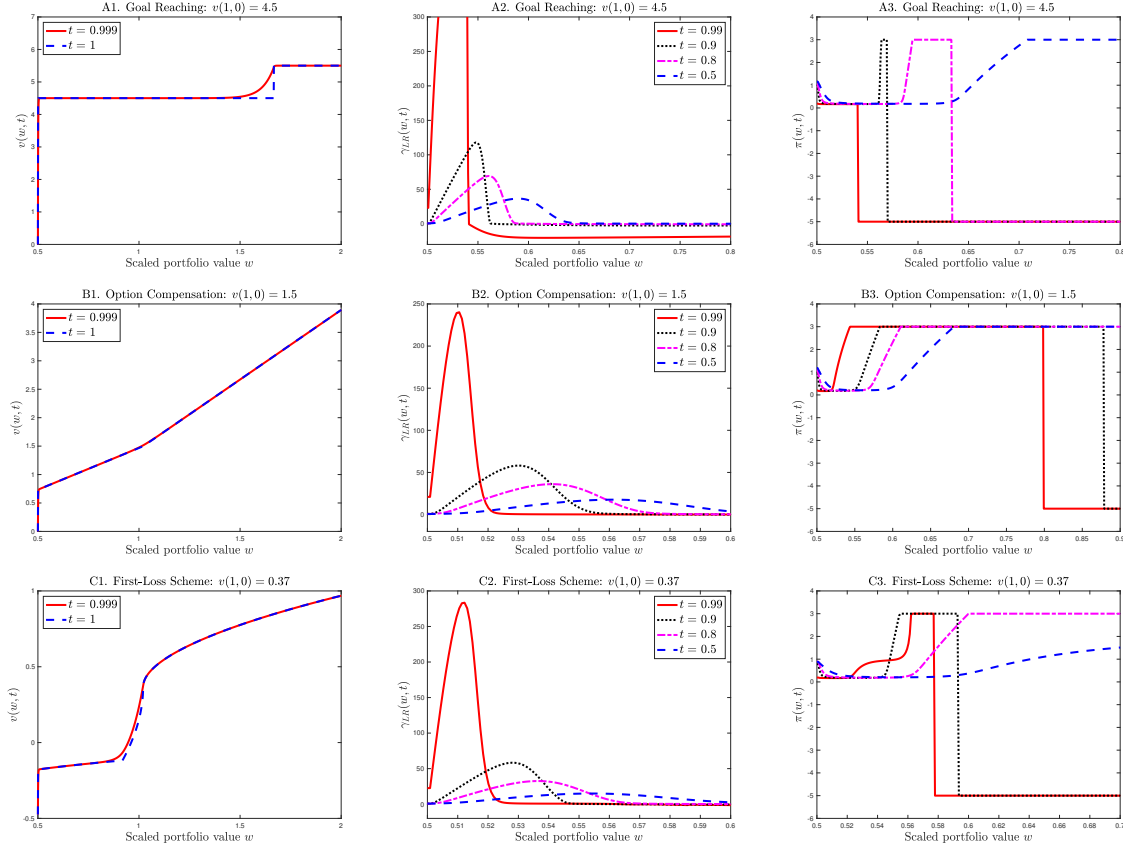


Figure 11: SCALED VALUE FUNCTION, LOCAL RISK AVERSION AND OPTIMAL INVESTMENT FOR OTHER NON-CONCAVE UTILITY FUNCTIONS. 1) Goal Reaching Problem with $\kappa = 0.6$: Panel A1 (resp. B1 and C1) plots the scaled value function $v(w, t)$, Panel A2 (resp. B2 and C2) plots the local risk aversion $\gamma_{LR}(w, t)$, and Panel A3 (resp. B3 and C3) plots the optimal investment strategy for goal reaching problem (resp. option compensation and first-loss scheme). For the goal reaching problem, we intentionally assume a large evaluation cost, $\kappa = 0.6$. For option compensation, due to the linear growth rate, we use a large discount rate $\delta = 0.4$. For the first-loss scheme, we assume that the proportions $\gamma = 0.1$ and $a = 0.2$ in (15), and the scaled utility $u(\cdot)$ is an S -shape given by (13). Other parameter values are listed in Table 2.

B Theoretical Analysis and Proofs

This section is devoted to the proof of Theorem 1. To this, we first consider a related finite-horizon portfolio selection problem with a given terminal objective function. After establishing some key estimates, we complete the proof of Theorem 1 by showing there is a unique fixed point with the help of the contraction mapping theorem.

We begin with the derivation of the equivalent characterization of the optimal control problem (17) by using model's homogeneity property and the change of measure.

B.1 Homogeneity and Change of Measure

Thanks to the homogeneity of the utility function and the GBM assumption, the value function is also homogeneous of degree β . As a result, the original stochastic control problem (17) is equivalent to the following one:

$$v(x, t) = \max_{\pi_t \in [\underline{\pi}, \bar{\pi}]} \tilde{\mathbb{E}}_t \left[e^{-\tilde{\delta}(T-t)} \left(u(\kappa w_{T-}) + (\kappa w_{T-})^\beta v(1, 0) \right) \mathbf{1}_{\{T < \tau\}} + e^{-\tilde{\delta}(\tau-t)} u(\kappa w_{\tau-}) \mathbf{1}_{\{\tau \leq T\}} \right] \quad (34)$$

subject to the dynamics of $w_t = W_t/B_t$:

$$\frac{dw_t}{w_t} = \left[(r - \mu_B + \sigma_B^2(1 - \beta)) + (\mu - r - \rho\sigma\sigma_B(1 - \beta))\pi_t \right] dt + \sigma\pi_t d\tilde{Z}_t - \sigma_B d\tilde{Z}_t^B \quad (35)$$

under the new probability $\tilde{\mathbb{P}}$ defined as

$$\frac{d\tilde{\mathbb{P}}_t}{d\mathbb{P}_t} = e^{-\frac{1}{2}\beta^2\sigma_B^2(T-t) + \beta\sigma_B\tilde{Z}_{T-t}^B} . \quad (36)$$

Here $\tilde{\delta}$ is the effective discount rate given by (23), $\tilde{Z}$ and $\tilde{Z}^B$ are two standard Brownian motions with the correlation ρ , and $\tilde{\mathbb{E}}_t$ is the conditional expectation on $w_t = w$ under $\tilde{\mathbb{P}}$. Indeed, from the definition (17), we have

$$\begin{aligned} v(w, t) &= B^{-\beta} V(W, B, t) \\ &= B^{-\beta} \max_{\pi_t \in [\underline{\pi}, \bar{\pi}]} \mathbb{E}_t \left[e^{-\delta(T-t)} \left(U(\kappa W_{T-}, B_{T-}) + V(\kappa W_{T-}, \kappa W_{T-}, 0) \right) \mathbf{1}_{\{T < \tau\}} \right. \\ &\quad \left. + e^{-\delta(\tau-t)} U(\kappa W_{\tau-}, B_{\tau-}) \mathbf{1}_{\{\tau \leq T\}} \right] \\ &= B^{-\beta} \max_{\pi_t \in [\underline{\pi}, \bar{\pi}]} \mathbb{E}_t \left[e^{-\delta(T-t)} B_{T-}^\beta \left(u(\kappa w_{T-}) + (\kappa w_{T-})^\beta v(1, 0) \right) \mathbf{1}_{\{T < \tau\}} \right. \\ &\quad \left. + e^{-\delta(\tau-t)} B_{\tau-}^\beta u(\kappa w_{\tau-}) \mathbf{1}_{\{\tau \leq T\}} \right] \\ &= \max_{\pi_t \in [\underline{\pi}, \bar{\pi}]} \mathbb{E}_t \left[e^{-\delta(T-t)} e^{\beta[(\mu_B - \frac{1}{2}\sigma_B^2)(T-t) + \sigma_B\tilde{Z}_{T-t}^B]} \left(u(\kappa w_{T-}) + (\kappa w_{T-})^\beta v(1, 0) \right) \mathbf{1}_{\{T < \tau\}} \right. \\ &\quad \left. + e^{-\delta(\tau-t)} e^{\beta[(\mu_B - \frac{1}{2}\sigma_B^2)(\tau-t) + \sigma_B\tilde{Z}_{\tau-t}^B]} u(\kappa w_{\tau-}) \mathbf{1}_{\{\tau \leq T\}} \right] \\ &= \max_{\pi_t \in [\underline{\pi}, \bar{\pi}]} \mathbb{E}_t \left[e^{-\tilde{\delta}(T-t)} e^{-\frac{1}{2}\beta^2\sigma_B^2(T-t) + \beta\sigma_B\tilde{Z}_{T-t}^B} \left(u(\kappa w_{T-}) + (\kappa w_{T-})^\beta v(1, 0) \right) \mathbf{1}_{\{T < \tau\}} \right. \\ &\quad \left. + e^{-\tilde{\delta}(\tau-t)} e^{-\frac{1}{2}\beta^2\sigma_B^2(\tau-t) + \beta\sigma_B\tilde{Z}_{\tau-t}^B} u(\kappa w_{\tau-}) \mathbf{1}_{\{\tau \leq T\}} \right] \\ &= \max_{\pi_t \in [\underline{\pi}, \bar{\pi}]} \tilde{\mathbb{E}}_t \left[e^{-\tilde{\delta}(T-t)} \left(u(\kappa w_{T-}) + (\kappa w_{T-})^\beta v(1, 0) \right) \mathbf{1}_{\{T < \tau\}} + e^{-\tilde{\delta}(\tau-t)} u(\kappa w_{\tau-}) \mathbf{1}_{\{\tau \leq T\}} \right] , \end{aligned}$$

where $\tilde{\mathbb{E}}$ is the expectation under the new probability measure $\tilde{\mathbb{P}}$ such that (36) holds.

Moreover, $\tilde{Z}^B$ and $\tilde{Z}$ satisfying, respectively,

$$d\tilde{Z}_t^B = dZ_t^B - \beta\sigma_B dt , \quad d\tilde{Z}_t = dZ_t - \beta\sigma_B\rho\pi_t dt ,$$

are Brownian motions under $\tilde{\mathbb{P}}$. Applying Itô's lemma to $w_t = W_t/B_t$, we obtain the dynamics of w_t given by (35).

We next consider a related stochastic control problem with a given terminal objective.

B.2 A Related Stochastic Control Problem

We replace the unknown value $v(1, 0)$ in (34) with a given $a \in \mathbb{R}$, and consider the following auxiliary stochastic control problem with a finite horizon:

$$\phi(x, t) = \sup_{\pi_t \in [\underline{\pi}, \bar{\pi}]} J(x, t; \pi_t), \quad (37)$$

subject to

$$\frac{dX_t}{X_t} = \left[(r - \mu_B + \sigma_B^2(1 - \beta)) + (\mu - r - \rho\sigma\sigma_B(1 - \beta))\pi_t \right] dt + \sigma\pi_t d\tilde{Z}_t - \sigma_B d\tilde{Z}_t^B. \quad (38)$$

Here the objective function is defined as

$$J(x, t; \pi_t) = \tilde{\mathbb{E}}_t \left[e^{-\tilde{\delta}(T-t)} U^a(X_T) \mathbf{1}_{\{T < \tau_x\}} + e^{-\tilde{\delta}(\tau_x-t)} u(\kappa X_{\tau_x}) \mathbf{1}_{\{T \geq \tau_x\}} \right] \quad (39)$$

where $U^a(\cdot)$ for $a \in \mathbb{R}$ is defined as

$$U^a(x) = u(\kappa x) + a x^\beta \mathbf{1}_{\{x > b\}} \quad \text{for } x \geq b, \quad (40)$$

$\tilde{\delta} = \delta - \beta\mu_B + \frac{1}{2}\sigma_B^2\beta(1 - \beta)$, and $\tau_x = \inf\{s \geq t \mid X_s \leq b, X_t = x \geq b\}$.

HJB Equation. By the dynamic programming principle, ϕ is expected to satisfy the following HJB equation

$$\frac{\partial \phi}{\partial t} + H(x, \phi, \phi_x, \phi_{xx}) = 0 \quad \text{for } x \in (b, \infty), t \in [0, T], \quad (41)$$

where the Hamiltonian $H(\cdot)$ is defined in (22).

As discussed in Section 3, following Dai et al. (2022) we impose the following terminal condition, for $x \geq b$:

$$\begin{cases} \lim_{(z,t) \rightarrow (x,T-)} \phi(z, t) - U^a(x-) \\ \quad - 2\Phi\left(\frac{\min\{0, \ln(z/x)\}}{L\sqrt{T-t}}\right) (U^a(x) - U^a(x-)) = 0 & \text{if } [\underline{\pi}, \bar{\pi}] \text{ is bounded,} \\ \lim_{(z,t) \rightarrow (x,T-)} \phi(z, t) = \hat{U}^a(x) & \text{if } [\underline{\pi}, \bar{\pi}] \text{ is unbounded,} \end{cases} \quad (42)$$

where constant $L = \sigma \max\{|\underline{\pi}|, |\bar{\pi}|\}$, $\Phi(\cdot)$ is the standard normal cumulative distribution function, and $\hat{U}^a(\cdot)$ is the concave envelope of $U^a(\cdot)$.

For the condition on the boundary $x = b$, we have two cases.

Case (a): $\sigma_B > 0$ or $\mu_B > r$. When the probability of going bankrupt is strictly positive, the Dirichlet-type boundary condition holds:

$$\phi(b, t) = u(\kappa b) \quad \text{for } t \in [0, T) , \quad (43)$$

Case (b): $\sigma_B = 0$ and $\mu_B \leq r$. When the probability of going bankrupt is zero for some admissible strategy ($\pi_t = 0$), the Neumann-type boundary condition holds:

$$\min \left\{ -\frac{\partial \phi}{\partial t} - (r - \mu_B)b \phi_x + (\delta - \beta \mu_B)\phi, \phi - u(\kappa b) \right\} = 0 \quad \text{for } t \in [0, T) . \quad (44)$$

Since the optimal decision holds for any $t < T$, which implies the following modification of the terminal value on the boundary $x = b$:

$$\lim_{(x,t) \rightarrow (b+, T-)} \phi(x, t) = \max\{U^a(b+), u(\kappa b)\} . \quad (45)$$

Next, we introduce the viscosity solution to these two cases, respectively. For easy exposition, we define the upper-semicontinuous envelope ϕ^* and lower-semicontinuous envelope ϕ_* in a domain $\mathcal{D}$ for a function ϕ , respectively, by

$$\phi^*(x) = \limsup_{x' \rightarrow x} \phi(x'), \quad \phi_*(x) = \liminf_{x' \rightarrow x} \phi(x') \quad \forall x \in \mathcal{D} . \quad (46)$$

B.3 Viscosity Solution to Case (a)

For Case (a), we introduce the following definition of viscosity solutions with Dirichlet boundary condition.

Definition 1 (Viscosity Solution with Dirichlet Boundary Condition for Case (a)). Suppose $\sigma_B > 0$ or $\mu_B > r$. We say ϕ is a viscosity subsolution of the HJB equation (41) with the terminal condition (42) and the Dirichlet boundary condition (43) if it satisfies the following conditions:

(i) For all smooth φ such that $\phi^* \leq \varphi$ and $\phi^*(\bar{x}, \bar{t}) = \varphi(\bar{x}, \bar{t})$ for some $(\bar{x}, \bar{t}) \in (b, \infty) \times [0, T)$,

$$-\frac{\partial \varphi}{\partial t}(\bar{x}, \bar{t}) - H^*(\bar{x}, \varphi(\bar{x}, \bar{t}), \varphi_x(\bar{x}, \bar{t}), \varphi_{xx}(\bar{x}, \bar{t})) \leq 0 .$$

(ii) For all $x \geq b$,

$$\begin{cases} \lim_{(z,t) \rightarrow (x, T-)} \phi^*(z, t) - U^a(x-) \\ \quad - 2\Phi\left(\frac{\min\{0, \ln(z/x)\}}{L\sqrt{T-t}}\right) \left(U^a(x) - U^a(x-)\right) \leq 0 & \text{if } [\underline{\pi}, \bar{\pi}] \text{ is bounded ,} \\ \lim_{(z,t) \rightarrow (x, T-)} \phi^*(z, t) \leq \widehat{U}^a(x) & \text{if } [\underline{\pi}, \bar{\pi}] \text{ is unbounded .} \end{cases}$$

(iii) For $t \in [0, T)$,

$$\phi^*(b, t) \leq u(\kappa b) .$$

We say ϕ is a viscosity supersolution of the HJB equation (41) with the terminal condition (42) and the Dirichlet boundary condition (43) if it satisfies the following conditions:

(i) For all smooth φ such that $\phi_* \geq \varphi$ and $\phi_*(\bar{x}, \bar{t}) = \varphi(\bar{x}, \bar{t})$ for some $(\bar{x}, \bar{t}) \in (b, \infty) \times [0, T)$,

$$-\frac{\partial \varphi}{\partial t}(\bar{x}, \bar{t}) - H_*(\bar{x}, \varphi(\bar{x}, \bar{t}), \varphi_x(\bar{x}, \bar{t}), \varphi_{xx}(\bar{x}, \bar{t})) \geq 0$$

(ii) For all $x \geq b$,

$$\begin{cases} \lim_{(z,t) \rightarrow (x,T-)} \phi_*(z, t) - U^a(x-) \\ \quad - 2\Phi\left(\frac{\min\{0, \ln(z/x)\}}{L\sqrt{T-t}}\right) (U^a(x) - U^a(x-)) \geq 0 & \text{if } [\underline{\pi}, \bar{\pi}] \text{ is bounded ,} \\ \lim_{(z,t) \rightarrow (x,T-)} \phi_*(z, t) \geq \widehat{U}^a(x) & \text{if } [\underline{\pi}, \bar{\pi}] \text{ is unbounded .} \end{cases}$$

(iii) For $t \in [0, T)$,

$$\phi_*(b, t) \geq u(\kappa b) .$$

Proposition 1 (Viscosity Solution with Dirichlet Boundary Condition). *Suppose $\sigma_B > 0$ or $\mu_B > r$. The HJB equation (41) coupled with the terminal condition (42) and the Dirichlet boundary condition (43) has a unique viscosity solution $\phi \in C([b, \infty) \times [0, T))$. In addition, if $\sigma_B = 0$, $\mu_B > r$, and the investment set is unbounded, $\phi \in C^{2,1}([b, \infty) \times [0, T))$.*

Proof. The first part follows from Theorem 4.2 in Dai et al. (2022). The regularity result follows from Theorem 4.2 in Bian et al. (2019). $\square$ $\square$

B.4 Viscosity Solution to Case (b)

For Case (b), we introduce the following viscosity solution with Neumann boundary condition.

Definition 2 (Viscosity Solution with Neumann Boundary Condition for Case (b)). *Suppose $\sigma_B = 0$ and $\mu_B \leq r$. We say ϕ is a viscosity subsolution of the HJB equation (41) with the terminal condition (42) and (45) and the Neumann boundary condition (44) if it satisfies conditions (i) and (ii) for viscosity subsolution in Definition 1, and the following condition:*

(iii) For all smooth φ such that $\phi^* \leq \varphi$ and $\phi^*(\bar{x}, \bar{t}) = \varphi(\bar{x}, \bar{t})$ for $\bar{x} = b$ and some $\bar{t} \in [0, T)$,

$$\min \left\{ -\frac{\partial \varphi}{\partial t}(\bar{x}, \bar{t}) - (r - \mu_B)b \varphi_x(\bar{x}, \bar{t}) + (\delta - \beta \mu_B)\varphi(\bar{x}, \bar{t}), \varphi(\bar{x}, \bar{t}) - u(\kappa b) \right\} \leq 0 .$$

We say ϕ is a viscosity supersolution of the HJB equation (41) with the terminal condition (42) and (45) and the Neumann boundary condition (44) if it satisfies conditions (i) and (ii) for viscosity supersolution in Definition 1, and the following condition:

(iii) For all smooth φ such that $\phi_* \geq \varphi$ and $\phi_*(\bar{x}, \bar{t}) = \varphi(\bar{x}, \bar{t})$ for $\bar{x} = b$ and some $\bar{t} \in [0, T)$,

$$\min \left\{ -\frac{\partial \varphi}{\partial t}(\bar{x}, \bar{t}) - (r - \mu_B)b \varphi_x(\bar{x}, \bar{t}) + (\delta - \beta \mu_B)\varphi(\bar{x}, \bar{t}), \varphi(\bar{x}, \bar{t}) - u(\kappa b) \right\} \geq 0 .$$

Proposition 2 (Viscosity Solution with Neumann Boundary Condition). *Suppose $\sigma_B = 0$ and $\mu_B \leq r$. The HJB equation (41) coupled with the terminal condition (42) and (45) and the Neumann boundary condition (44) has a unique viscosity solution $\phi \in C([b, \infty) \times [0, T))$. In addition, if the investment set is unbounded, $\phi \in C^{2,1}([b, \infty) \times [0, T))$.*

Remark B.1. When $\sigma_B = 0$ and $\mu_B = r$, the boundary condition (44) simplifies to

$$\min \left\{ -\frac{\partial \phi}{\partial t} + (\delta - \beta \mu_B)\phi, \phi - u(\kappa b) \right\} = 0 \quad \text{for } t \in [0, T) .$$

This equation coupled with the terminal condition (42) yields a closed-form solution:

$$\phi(b, t) = \max\{U^a(b+), u(\kappa b)\} e^{-(\delta - \beta \mu_B)(T-t)} \quad \text{for } t \in [0, T]. \quad (47)$$

So, one can equivalently use viscosity solution with Dirichlet boundary condition (47) for this special case.

Interestingly, this special case is also equivalent to a state constrained problem because it is never optimal to do liquidation before the maturity even when $a < 0$. To see this, note that $\max\{U^a(b+), u(\kappa b)\} < 0$ for all $a < 0$, therefore, postponing the liquidation is always preferable. As a result, one can alternatively use state constrained viscosity solutions to handle the boundary condition; see [Soner \(1986a,b\)](#); [Altarovici et al. \(2017\)](#).

The viscosity property in the interior domain is standard, so we only give the proof for the boundary condition. The proof of Proposition 2 is divided into the following lemmas.

Lemma B.1 (Supersolution). *The value function $\phi_*(b, \cdot)$ is a viscosity supersolution of (44) on the boundary $x = b$.*

Proof. Fix $t < T$, and take some arbitrary sequence $(x_m, t_m) \rightarrow (b, t)$ with $x_m > b$ and $t_m < T$. Fix a nonzero constant control $\pi \in [\underline{\pi}, \bar{\pi}]$ and denote by $X_u^{x_m, t_m, \pi}$ the associated process for $u \geq t_m$. By continuity, it is clear that

$$X_{\tau_{x_m}}^{x_m, t_m, \pi} = b, \quad \liminf_{m \rightarrow \infty} \tilde{\mathbb{P}}(\tau_{x_m} < T) = 1 ,$$

where τ_x denotes the first hitting time to the boundary b starting from x . Under the lower-bound condition on $U(x, 1)$, we can apply Fatou's lemma to obtain

$$\phi_*(b, t) = \liminf_{m \rightarrow \infty} \phi(x_m, t_m) \geq \tilde{\mathbb{E}}[\liminf_{m \rightarrow \infty} e^{-\tilde{\delta}(\tau_{x_m} - t_m)} u(\kappa X_{\tau_{x_m}}^{x_m, t_m, \pi}) \mathbf{1}_{\{\tau_{x_m} < T\}}] = u(\kappa b) .$$

On the other hand, take $\pi \equiv 0 \in [\underline{\pi}, \bar{\pi}]$. It is clear that

$$\tilde{\mathbb{P}}(\tau_{x_m} < T) = 0 .$$

Fix $t_\varepsilon \in (t_m, T)$, by dynamic programming principle,

$$\phi(x_m, t_m) \geq \tilde{\mathbb{E}}[e^{-\tilde{\delta}(t_\varepsilon - t_m)} \phi(X_{t_\varepsilon}^{x_m, t_m, \pi}, t_\varepsilon)] \geq \tilde{\mathbb{E}}[e^{-\tilde{\delta}(t_\varepsilon - t_m)} \varphi(X_{t_\varepsilon}^{x_m, t_m, \pi}, t_\varepsilon)] ,$$

where the second inequality follows from the fact that $\phi \geq \varphi$. Applying Itô's lemma to $e^{-\delta(u - t_m)} \varphi(X_u, u)$ between t_m and t_ε , and sending $t_\varepsilon \rightarrow t_m$, we immediately have

$$-\frac{\partial \varphi}{\partial t}(x_m, t_m) - (r - \mu_B) x_m \varphi_x(x_m, t_m) + (\delta - \beta \mu_B) \varphi(x_m, t_m) \geq 0 .$$

Sending $m \rightarrow \infty$, we obtain the required inequality. □ □

Lemma B.2 (Subsolution). *The value function $\phi^*(b, \cdot)$ is a viscosity subsolution of (44) on the boundary $x = b$.*

Proof. We argue by contradiction assuming that

$$\min \left\{ -\frac{\partial \varphi}{\partial t}(b, t) - (r - \mu_B) b \varphi_x(b, t) + (\delta - \beta \mu_B) \varphi(b, t), \varphi(b, t) - u(\kappa b) \right\} \geq \varepsilon > 0$$

for some test function φ . By continuity, there exists a positive η such that for $(x_m, t_m) \in (b, b + \eta) \times (t - \eta, t + \eta)$:

$$\begin{aligned} -\frac{\partial \varphi}{\partial t}(x_m, t_m) - (r - \mu_B) x_m \varphi_x(x_m, t_m) + (\delta - \beta \mu_B) \varphi(x_m, t_m) &\geq \varepsilon , \\ \varphi(x_m, t_m) - u(\kappa x_m) &\geq \varepsilon . \end{aligned}$$

By definition of ϕ^* , we have

$$\gamma_m = \phi(x_m, t_m) - \varphi(x_m, t_m) \rightarrow 0 \quad \text{as } m \rightarrow \infty$$

For later use, let $h_m > 0$ such that $h_m \rightarrow 0$ and $h_m/\gamma_m \rightarrow 0$ as $m \rightarrow \infty$.

From dynamical programming principle, e.g., equation (3.19) in [Pham \(2009\)](#), there exists an ε -optimal strategy π_ε such that

$$\phi(x_m, t_m) - \frac{h_m \varepsilon}{4} \leq \tilde{\mathbb{E}} \left[e^{-\tilde{\delta}(\tau - t_m)} \phi(X_\tau^{\pi_\varepsilon}, \tau) \right] \quad (48)$$

for any stopping time $\tau \in (t_m, T)$.

We next modify π_ε as

$$\pi_{\varepsilon,0}(s) = \begin{cases} 0 & \text{when } s < \tau_m \\ \pi_\varepsilon(s) & \text{when } s \geq \tau_m \end{cases} \quad (49)$$

where τ_m is the first exit time defined as

$$\tau_m = \inf \{s \in (t_m, t_m + \min\{\eta, h_m\}) : X_u \notin (b, (1 + \varepsilon_m)x_m) \subset (b, b + \eta)\}$$

with $\varepsilon_m > 0$ to be chosen later. Since $\varphi(x_m, t_m) - U(\kappa x_m, 1) > 0$ and $\pi = 0$ is the only admissible strategy to make the agent stay alive, $\pi_{\varepsilon,0}$ is an ε -optimal strategy such that

$$\phi(x_m, t_m) - \frac{h_m \varepsilon}{2} \leq \tilde{\mathbb{E}} \left[e^{-\tilde{\delta}(\tau - t_m)} \phi(X_\tau^{\pi_{\varepsilon,0}}, \tau) \right], \quad (50)$$

when x_m is sufficiently close to b .

Using $\pi_{\varepsilon,0}$ and taking $\tau = \tau_m$, we can obtain

$$\begin{aligned} \varphi(x_m, t_m) + \gamma_m - \frac{h_m \varepsilon}{2} &\leq \tilde{\mathbb{E}} \left[e^{-\tilde{\delta}(\tau_m - t_m)} \phi(X_{\tau_m}^{x_m, t_m, \pi_{\varepsilon,0}}, \tau_m) \right] \\ &\leq \tilde{\mathbb{E}} \left[e^{-\tilde{\delta}(\tau_m - t_m)} \varphi(X_{\tau_m}^{x_m, t_m, \pi_{\varepsilon,0}}, \tau_m) \right]. \end{aligned}$$

Applying Itô's lemma to $e^{-\tilde{\delta}(s - t_m)} \varphi(X_s^{\pi_{\varepsilon,0}}, s)$ between t_m and τ_m , we have

$$\begin{aligned} 0 &\geq \frac{\gamma_m}{h_m} - \frac{\varepsilon}{2} + \tilde{\mathbb{E}} \left[\frac{1}{h_m} \int_{t_m}^{\tau_m} e^{-\tilde{\delta}(s - t_m)} \left(\varphi_t - (r - \mu_B)x\varphi_x + (\tilde{\delta} - \beta\mu_B)\varphi \right) ds \right] \\ &\geq \frac{\gamma_m}{h_m} + \varepsilon \left(\frac{1}{h_m} \tilde{\mathbb{E}}[\tau_m - t_m] - \frac{1}{2} + o(h_m) \right), \end{aligned}$$

where the second inequality follows from the fact that $\varphi_t - (r - \mu_B)x\varphi_x + (\delta - \beta\mu_B)\varphi \geq \varepsilon$. Choosing $\varepsilon_m = e^{(r - \mu_B)h_m} - 1 > 0$, then $\frac{1}{h_m} \tilde{\mathbb{E}}[\tau_m - t_m] \rightarrow 1$ as $m \rightarrow \infty$. Therefore, by sending $m \rightarrow \infty$, we obtain a contradiction that $0 \geq \varepsilon/2 > 0$. This completes the proof. $\square$ $\square$

Lemma B.3 (Comparison Principle). *Let $\underline{\phi}$ and $\bar{\phi}$ be respectively the viscosity sub- and supersolution of the HJB equation (41) with the terminal condition (42) and (45) and the Neumann boundary condition (44). Then $\underline{\phi} \leq \bar{\phi}$.*

Proof. We use a contraposition argument by assuming that

$$M = \sup_{(b,\infty) \times [0,T)} \{\underline{\phi} - \bar{\phi}\} > 0 .$$

If the maximum is attained in the interior point, then the classic argument applies immediately, see [Crandall et al. \(1992\)](#). Hence, we only consider the case that the maximum is attained on the boundary (b, t) for some $t \in [0, T)$.

First, by the definition, we have

$$\begin{aligned} \min \left\{ -\frac{\partial \bar{\varphi}}{\partial t}(b, t) - (r - \mu_B)b \bar{\varphi}_x(b, t) + (\delta - \beta\mu_B)\bar{\varphi}(b, t), \bar{\varphi}(b, t) - u(\kappa b) \right\} &\geq 0, \\ \min \left\{ -\frac{\partial \underline{\varphi}}{\partial t}(b, t) - (r - \mu_B)b \underline{\varphi}_x(b, t) + (\delta - \beta\mu_B)\underline{\varphi}(b, t), \underline{\varphi}(b, t) - u(\kappa b) \right\} &\leq 0, \end{aligned}$$

where $\bar{\varphi}$ and $\underline{\varphi}$ are test functions for viscosity super- and subsolution, respectively.

By assumption, $\underline{\varphi}(b, t) - \bar{\varphi}(b, t) = M > 0$. On the other hand, the supersolution property implies that

$$\underline{\varphi}(b, t) > \bar{\varphi}(b, t) \geq u(\kappa b) .$$

Then the subsolution property yields that

$$-\frac{\partial \underline{\varphi}}{\partial t}(b, t) - (r - \mu_B)b \underline{\varphi}_x(b, t) + (\delta - \beta\mu_B)\underline{\varphi}(b, t) \leq 0 . \quad (51)$$

Meanwhile, the supersolution also implies

$$-\frac{\partial \bar{\varphi}}{\partial t}(b, t) - (r - \mu_B)b \bar{\varphi}_x(b, t) + (\delta - \beta\mu_B)\bar{\varphi}(b, t) \geq 0 . \quad (52)$$

Combining (51) and (52) and defining $\varphi = \underline{\varphi} - \bar{\varphi}$, we have

$$0 \geq -\frac{\partial \varphi}{\partial t}(b, t) - (r - \mu_B)b \varphi_x(b, t) + (\delta - \beta\mu_B)\varphi(b, t) .$$

Finally, since (b, t) is a maximum point, $\frac{\partial \varphi}{\partial t}(b, t) \leq 0$ and $\varphi_x(b, t) \leq 0$. Using two inequalities, we obtain a contradiction that

$$0 \geq (\delta - \beta\mu_B)\varphi(b, t) = (\delta - \beta\mu_B)M > 0 ,$$

which completes the proof. □ □

Remark B.2. By the comparison principle, we have $\phi_* \geq \phi^*$. On the other hand, $\phi_* \leq \phi^*$ from the definition. Consequently, $\phi = \phi_* = \phi^*$, which leads to the continuity of viscosity solution to the HJB equation (41) with the Neumann boundary condition (44) on the boundary.

Next, we provide a key estimate, which is critical to the construction of a contraction mapping.

B.5 A Key Estimate

Lemma B.4. *Let the condition (24) hold. For $a_1 < a_2$, let ϕ_1 and ϕ_2 be the unique viscosity solutions to the problem (41) with $a = a_1$ and $a = a_2$, respectively. Then*

$$0 \leq \phi_2(x, t) - \phi_1(x, t) \leq (a_2 - a_1) x^{\hat{\beta}} e^{-\hat{\delta}(T-t)} , \quad (53)$$

where $\hat{\beta} = \max\{\alpha, \beta\} \leq 1$, and

$$\begin{aligned} 0 < \hat{\delta} < \delta - \mu_B \beta - (r - \mu_B) \hat{\beta} - \max \left\{ 0, \frac{1}{2} \sigma_B^2 \beta (\beta - 1) + \frac{1}{2} \sigma^2 \hat{\beta} (\hat{\beta} - 1) + \sigma_B^2 \hat{\beta} (1 - \beta) \right. \\ \left. - \max_{\pi \in [\underline{\pi}, \bar{\pi}]} \left\{ \frac{1}{2} \sigma^2 \hat{\beta} (\hat{\beta} - 1) \pi^2 + [\mu - r - \rho \sigma \sigma_B (\hat{\beta} - \beta)] \hat{\beta} \pi \right\} \right\} . \end{aligned} \quad (54)$$

Proof. Since the terminal condition is monotonically increasing with respect to a , it is easy to see that $\phi_2 \geq \phi_1$. For the second inequality, we need to show that

$$\bar{\phi}(x, t) := \phi_1(x, t) + (a_2 - a_1) x^{\hat{\beta}} e^{-\hat{\delta}(T-t)}$$

is a supersolution to the problem (41) with $a = a_2$. Let's first consider the Dirichlet case of Definition 1. That is, the following conditions hold:

- (a) For all smooth φ such that $\bar{\phi}_* \geq \varphi$ and $\bar{\phi}_*(\bar{x}, \bar{t}) = \varphi(\bar{x}, \bar{t})$ for some $(\bar{x}, \bar{t}) \in [0, T) \times (b, \infty)$,

$$-\frac{\partial \varphi}{\partial t}(\bar{x}, \bar{t}) - H(\bar{x}, \varphi(\bar{x}, \bar{t}), \varphi_x(\bar{x}, \bar{t}), \varphi_{xx}(\bar{x}, \bar{t})) \geq 0 . \quad (55)$$

- (b) For all $0 \leq t \leq T$,

$$\lim_{(z, s) \rightarrow (b, t)} \bar{\phi}(z, s) - u(\kappa b) \geq 0 . \quad (56)$$

- (c) For all $x > b$,

$$\lim_{(z, t) \rightarrow (x, T)} \bar{\phi}(z, t) - K^{a_2}(z-) - 2\Phi\left(\frac{\min\{0, \ln(z/x)\}}{L\sqrt{T-t}}\right) (K^{a_2}(x) - K^{a_2}(x-)) \geq 0 . \quad (57)$$

where $K^a(x) = U^a(x)$ if $[\underline{\pi}, \bar{\pi}]$ is bounded, and $K^a(x) = \hat{U}^a(x)$ otherwise.

First, for the boundary condition (56), we have

$$\begin{aligned} \lim_{(x, s) \rightarrow (b, t)} \bar{\phi}(x, \tau) - u(\kappa b) &= \lim_{(x, s) \rightarrow (b, t)} \phi_1(x, s) + (a_2 - a_1) x^{\hat{\beta}} e^{-\hat{\delta}(T-s)} - u(\kappa b) \\ &\geq \lim_{(x, s) \rightarrow (b, t)} \phi_1(x, s) - u(\kappa b) \geq 0 , \end{aligned}$$

since ϕ_1 is a viscosity solution when $a = a_1$.

Next, for the terminal condition (57), note first that the definition of $U^a(\cdot)$ implies

$$K^{a_2}(x) - K^{a_1}(x) = K^{a_2}(x-) - K^{a_1}(x-) = (a_2 - a_1)x^{\hat{\beta}}\mathbf{1}_{\{x>b\}}.$$

As a result,

$$\begin{aligned} & \lim_{(z,t) \rightarrow (x,T)} \bar{\phi}(z,t) - K^{a_2}(x-) - 2\Phi\left(\frac{\min\{0, \ln(z/x)\}}{L\sqrt{T-t}}\right) \left(K^{a_2}(x) - K^{a_2}(x-)\right) \\ &= \lim_{(z,t) \rightarrow (x,T)} \phi_1(z,t) - K^{a_1}(x-) - 2\Phi\left(\frac{\min\{0, \ln(z/x)\}}{L\sqrt{T-t}}\right) \left(K^{a_1}(x) - K^{a_1}(x-)\right) \\ & \quad + (a_2 - a_1)z^{\hat{\beta}}e^{-\hat{\delta}(T-t)} - \left(K^{a_2}(x-) - K^{a_1}(x-)\right) \\ & \quad - 2\Phi\left(\frac{\min\{0, \ln(z/x)\}}{L\sqrt{T-t}}\right) \left(K^{a_2}(x) - K^{a_2}(x-) - K^{a_1}(x) + K^{a_1}(x-)\right) \\ & \geq \lim_{(z,t) \rightarrow (x,T)} (a_2 - a_1)z^{\hat{\beta}}e^{-\hat{\delta}(T-t)} - \left(K^{a_2}(x-) - K^{a_1}(x-)\right) = 0, \end{aligned}$$

since ϕ_1 is a viscosity solution when $a = a_1$.

Now, let's check the inequality (55). To this end, note that the smooth test function φ can always be written as $\varphi = \psi + (a_2 - a_1)x^{\hat{\beta}}e^{-\hat{\delta}(T-t)}$, where ψ is a smooth test function for ϕ_1 . Consequently, by denoting $w(x,t) = (a_2 - a_1)x^{\hat{\beta}}e^{-\hat{\delta}(T-t)}$, we have

$$\begin{aligned} & -\frac{\partial \varphi}{\partial t}(\bar{x}, \bar{t}) - H(\bar{x}, \varphi(\bar{x}, \bar{t}), \varphi_x(\bar{x}, \bar{t}), \varphi_{xx}(\bar{x}, \bar{t})) \\ & \geq -\frac{\partial \psi}{\partial t}(\bar{x}, \bar{t}) - H(\bar{x}, \psi(\bar{x}, \bar{t}), \psi_x(\bar{x}, \bar{t}), \psi_{xx}(\bar{x}, \bar{t})) \\ & \quad -\frac{\partial w}{\partial t}(\bar{x}, \bar{t}) - H(\bar{x}, w(\bar{x}, \bar{t}), w_x(\bar{x}, \bar{t}), w_{xx}(\bar{x}, \bar{t})) \\ & \geq -\frac{\partial w}{\partial t}(\bar{x}, \bar{t}) - H(\bar{x}, w(\bar{x}, \bar{t}), w_x(\bar{x}, \bar{t}), w_{xx}(\bar{x}, \bar{t})) \\ & = \left(\delta - \hat{\delta} + \frac{1}{2}\sigma_B^2\beta(1-\beta) - \mu_B\beta - \frac{1}{2}\sigma^2\hat{\beta}(\hat{\beta}-1) - [r - \mu_B + (1-\beta)\sigma_B^2]\hat{\beta} \right. \\ & \quad \left. - \max_{\pi \in [\underline{\pi}, \bar{\pi}]} \left\{ \frac{1}{2}\sigma^2\hat{\beta}(\hat{\beta}-1)\pi^2 + [\mu - r - \rho\sigma\sigma_B(\hat{\beta}-\beta)]\hat{\beta}\pi \right\} \right) w(\bar{x}, \bar{t}) \geq 0 \end{aligned}$$

by using the condition of $\hat{\delta}$. Therefore, $\bar{\phi}$ is a supersolution for $a = a_2$ for the Dirichlet case.

Finally, for the Neumann case of Definition 2, we only need to check the boundary condition (44), i.e.,

$$\min \left\{ -\frac{\partial \bar{\phi}}{\partial t} - (r - \mu_B)b\bar{\phi}_x + (\delta - \beta\mu_B)\bar{\phi}, \bar{\phi} - u(\kappa b) \right\} \geq 0 \quad \text{for } t \in [0, T].$$

To this end, we use the superviscosity property of ϕ_1 to yield that

$$\bar{\phi} - u(\kappa b) = \phi_1 + (a_2 - a_1)b^{\hat{\beta}}e^{-\hat{\delta}(T-t)} - u(\kappa b) \geq \phi_1 - u(\kappa b) \geq 0$$

on one hand, and that

$$\begin{aligned}
-\frac{\partial \bar{\phi}}{\partial t} - (r - \mu_B)b\bar{\phi}_x + (\delta - \beta\mu_B)\bar{\phi} &= -\frac{\partial \phi_1}{\partial t} - (r - \mu_B)b\phi_{1,x} + (\delta - \beta\mu_B)\phi_1 \\
&\quad + (\delta - \hat{\delta} - \beta\mu_B - \hat{\beta}(r - \mu_B))(a_2 - a_1)b^{\hat{\beta}}e^{-\hat{\delta}(T-t)} \\
&\geq -\frac{\partial \phi_1}{\partial t} - (r - \mu_B)b\phi_{1,x} + (\delta - \hat{\beta}\mu_B)\phi_1 \geq 0
\end{aligned}$$

on the other hand. This completes the proof. $\square$ $\square$

Now let us prove Theorem 1 by a fixed point argument.

B.6 Proof of Theorem 1

Proof. Define a normed space $\mathbf{X}$ and its associated norm $\|\cdot\|$ respectively by

$$\mathbf{X} = \left\{ \phi \in C([b, \infty) \times [0, T)) \right\}, \quad (58)$$

$$\|\phi\| = \sup_{(x,t) \in [b, \infty) \times [0, T)} \left| \frac{e^{-\hat{\delta}t}}{\max(x, 1)} \phi(x, t) \right|, \quad (59)$$

where $\hat{\delta}$ satisfies (54). We claim that $(\mathbf{X}, \|\cdot\|)$ is complete; that is, for any Cauchy sequence $\{\phi_n\}_{n \geq 1}$ in $\mathbf{X}$, there exists $\phi \in \mathbf{X}$ such that $\lim_{n \rightarrow \infty} \|\phi_n - \phi\| = 0$.

To this end, for each $M > 0$, define a normed space $(\mathbf{X}^M, \|\cdot\|^M)$ as

$$\begin{aligned}
\mathbf{X}^M &= \left\{ \phi \in C([b, M] \times [0, T - 1/M]) \right\}, \\
\|\phi\|^M &= \sup_{(x,t) \in [b, M] \times [0, T - 1/M]} \left| \frac{e^{-\hat{\delta}t}}{\max(x, 1)} \phi(x, t) \right|,
\end{aligned}$$

which is complete. Next, define a sequence $\{\phi_n^M\}_{n \geq 1}$ in $\mathbf{X}^M$ with $\phi_n^M(x, t) = \phi_n(x, t)$ for $(x, t) \in [b, M] \times [0, T - 1/M]$. Since $\mathbf{X}^M$ is complete, ϕ_n^M admits a limit $\phi^M \in \mathbf{X}^M$. In addition, by the construction, it is easy to see that $\phi(x, t) = \lim_{M \rightarrow \infty} \phi^M(x, t)$ is well defined and continuous in $[b, \infty) \times [0, T)$, and $\phi_n(x, t)$ converges to $\phi(x, t)$ for every $(x, t) \in [b, \infty) \times [0, T)$.

It remains to show that $\phi \in \mathbf{X}$ and $\lim_{n \rightarrow \infty} \|\phi_n - \phi\| = 0$. For the former, we fix any $M > 0$, and use the fact that ϕ_n is a Cauchy sequence to obtain

$$\|\phi^M\|^M = \lim_{n \rightarrow \infty} \|\phi_n^M\|^M \leq \lim_{n \rightarrow \infty} \|\phi_n^M\| \leq \liminf_{n \rightarrow \infty} \|\phi_n\| \leq C,$$

for some positive constant C . Consequently, $\|\phi\| = \lim_{M \rightarrow \infty} \|\phi^M\|^M \leq C < \infty$, which implies that $\phi \in \mathbf{X}$.

For the latter, we choose $\varepsilon > 0$ and N such that $\|\phi_n - \phi_m\| < \varepsilon$ when $m, n > N$. For any $(x, t) \in [b, \infty) \times [0, T)$ and $n > N$,

$$\frac{e^{-\hat{\delta}t}}{\max(x, 1)} |\phi_n(x, t) - \phi(x, t)| = \lim_{m \rightarrow \infty} \frac{e^{-\hat{\delta}t}}{\max(x, 1)} |\phi_n(x, t) - \phi_m(x, t)| \leq \varepsilon,$$

where the equality follows from $\phi_m(x, t)$ converges to $\phi(x, t)$ for each $(x, t) \in (b, \infty) \times [0, T)$. Hence, by definition $\|\phi_n - \phi\| < \varepsilon$ for $n > N$. Consequently, $\lim_{n \rightarrow \infty} \|\phi_n - \phi\| = 0$.

Next, we turn to constructing a contraction mapping in $\mathbf{X}$. Define a mapping $\mathcal{F} : \phi \in \mathbf{X} \mapsto \mathbf{X}$ as the unique viscosity solution $\mathcal{F}[\phi]$ of the problem (41)-(42) with $a = \phi(1, 0)\kappa^\beta$. Then, we claim that $\mathcal{F}$ is a contraction. To see this, suppose $\phi_1 \in \mathbf{X}$ and $\phi_2 \in \mathbf{X}$, then by (53)

$$\begin{aligned} \|\mathcal{F}[\phi_1] - \mathcal{F}[\phi_2]\| &= \sup_{(x, t) \in [b, \infty) \times [0, T)} \left| \frac{e^{-\hat{\delta}t}}{\max(x, 1)} (\mathcal{F}[\phi_1] - \mathcal{F}[\phi_2]) \right| \\ &\leq \sup_{(x, t) \in [b, \infty) \times [0, T)} \left| \frac{e^{-\hat{\delta}t}}{\max(x, 1)} (\phi_1(1, 0) - \phi_2(1, 0)) \kappa^\beta x^{\hat{\beta}} e^{-\hat{\delta}(T-t)} \right| \\ &= e^{-\hat{\delta}T} \kappa^\beta \left| \phi_1(1, 0) - \phi_2(1, 0) \right| \\ &\leq e^{-\hat{\delta}T} \kappa^\beta \|\phi_1 - \phi_2\|, \end{aligned}$$

which is a contraction as $e^{-\hat{\delta}T} \kappa^\beta < 1$.

Finally, the regularity result follows from Propositions 1 and 2. This completes the proof of Theorem 1. □

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